

Homework 7: Neural Networks

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Setup

Install Required Packages

```
In [1]: !pip install -q torchmetrics torchinfo pytorch_lightning ISLP
```

```
Preparing metadata (setup.py) ... done
===== 983.4/983.4 kB 37.4 MB/s eta 0:00:00
===== 857.3/857.3 kB 28.0 MB/s eta 0:00:00
===== 16.6/16.6 MB 56.5 MB/s eta 0:00:00
===== 409.1/409.1 kB 21.4 MB/s eta 0:00:00
===== 84.6/84.6 kB 5.1 MB/s eta 0:00:00
===== 117.3/117.3 kB 10.4 MB/s eta 0:00:00
Building wheel for autograd-gamma (setup.py) ... done
```

```
In [2]: # Mount Google Drive to persist logs across runtime disconnects
from google.colab import drive
drive.mount('/content/drive')
```

```
LOG_DIR = '/content/drive/MyDrive/hw7_logs'
import os; os.makedirs(LOG_DIR, exist_ok=True)
print(f"Logs will be saved to: {LOG_DIR}")
```

Mounted at /content/drive
Logs will be saved to: /content/drive/MyDrive/hw7_logs

Imports

```
In [3]: import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

import torch
import torchvision
from torch import nn
from torch.optim import Adam
from torch.utils.data import TensorDataset, DataLoader, random_split
from pytorch_lightning.callbacks import Callback

import pytorch_lightning as pl
from torchinfo import summary
from torchvision import transforms, datasets
from pytorch_lightning import Trainer, seed_everything
from pytorch_lightning.loggers import CSVLogger
from pytorch_lightning.callbacks import TQDMProgressBar, ModelCheckpoint

from torchvision.datasets import CIFAR100
from ISLP.torch import SimpleDataModule, SimpleModule, ErrorTracker, rec_num_workers

seed_everything(0, workers=True)
torch.use_deterministic_algorithms(True, warn_only=True)
```

INFO:lightning_fabric.utilities.seed:Seed set to 0

Device Setup

```
In [4]: device = "cuda" if torch.cuda.is_available() else "cpu"
print(f"Using device: {device}")
max_num_workers = rec_num_workers()
print(f"Number of workers: {max_num_workers}")
```

Using device: cuda
Number of workers: 2

PrintMetrics Callback

This callback prints a summary line at the end of each epoch so you can track progress without scrolling through the progress bar.

```
In [5]: class PrintMetrics(Callback):
    def on_validation_epoch_end(self, trainer, pl_module):
        metrics = trainer.callback_metrics
        epoch = trainer.current_epoch
        train_loss = metrics.get("train_loss")
        valid_loss = metrics.get("valid_loss")
        valid_acc = metrics.get("valid_accuracy")
        train_acc = metrics.get("train_accuracy_epoch") or metrics.get("train_accuracy")
        lr = trainer.optimizers[0].param_groups[0]["lr"]
        if all(m is not None for m in [train_loss, train_acc, valid_loss, valid_acc]):
            print(
                f"Epoch {epoch:3d} | "
                f"train_loss: {train_loss.item():.4f} | "
                f"train_acc: {train_acc.item():.4f} | "
                f"valid_loss: {valid_loss.item():.4f} | "
                f"valid_acc: {valid_acc.item():.4f} | "
                f"lr: {lr:.6f}"
            )
```

CIFAR100 Normalization Constants

```
In [6]: cifar100_mean = (0.5071, 0.4865, 0.4409)
        cifar100_std = (0.2673, 0.2564, 0.2762)
```

Helper: Read Epoch Metrics from CSV Log

```
In [7]: def read_epoch_metrics(logger):
    path = os.path.join(logger.log_dir, "metrics.csv") # saved to Drive
    df = pd.read_csv(path)
    return df.groupby("epoch").agg(
        lambda x: x.dropna().iloc[-1] if x.dropna().size > 0 else np.nan
    )

def plot_accuracy(epoch_metrics, title):
    fig, ax = plt.subplots(figsize=(9, 5))
    ax.plot(epoch_metrics.index, epoch_metrics["train_accuracy_epoch"], label="Training accuracy")
    ax.plot(epoch_metrics.index, epoch_metrics["valid_accuracy"], label="Validation accuracy")
    ax.set_xlabel("Epoch")
    ax.set_ylabel("Accuracy")
    ax.set_title(title)
    ax.legend()
    ax.grid(True)
    plt.tight_layout()
    plt.show()

def unnormalize_cifar100(img):
    mean = torch.tensor(cifar100_mean).view(3, 1, 1)
    std = torch.tensor(cifar100_std).view(3, 1, 1)
    return torch.clamp(img * std + mean, 0, 1)
```

Model Building Blocks

```
In [8]: class BuildingBlock(nn.Module):
    """Single Conv -> BN -> ReLU -> MaxPool block (used in P1-P3)."""
    def __init__(self, in_channels, out_channels):
        super().__init__()
        self.conv = nn.Conv2d(in_channels, out_channels, kernel_size=3, padding="same")
        self.bn = nn.BatchNorm2d(out_channels)
        self.activation = nn.ReLU()
        self.pool = nn.MaxPool2d(kernel_size=2)

    def forward(self, x):
        return self.pool(self.activation(self.bn(self.conv(x))))

class DeepBlockNoBN(nn.Module):
    """Stacks num_convs (Conv->ReLU) layers before MaxPool. No BatchNorm (P4a)."""
    def __init__(self, in_channels, out_channels, num_convs=2):
        super().__init__()
```

```

        layers = [nn.Conv2d(in_channels, out_channels, kernel_size=3, padding=1), nn.ReLU()]
        for _ in range(num_convs - 1):
            layers += [nn.Conv2d(out_channels, out_channels, kernel_size=3, padding=1), nn.ReLU()]
        layers.append(nn.MaxPool2d(kernel_size=2))
        self.block = nn.Sequential(*layers)

    def forward(self, x):
        return self.block(x)

class DeepBlockBN(nn.Module):
    """Stacks num_convs (Conv->BN->ReLU) layers before MaxPool (P4b)."""
    def __init__(self, in_channels, out_channels, num_convs=2):
        super().__init__()
        layers = [nn.Conv2d(in_channels, out_channels, kernel_size=3, padding=1),
                  nn.BatchNorm2d(out_channels), nn.ReLU()]
        for _ in range(num_convs - 1):
            layers += [nn.Conv2d(out_channels, out_channels, kernel_size=3, padding=1),
                      nn.BatchNorm2d(out_channels), nn.ReLU()]
        layers.append(nn.MaxPool2d(kernel_size=2))
        self.block = nn.Sequential(*layers)

    def forward(self, x):
        return self.block(x)

```

Model Architectures

```

In [9]: class CIFARModel(nn.Module):
        """Original ISLP CNN: 3->32->64->128->256 channels, MLP 1024->512->100."""
        def __init__(self):
            super().__init__()
            self.conv = nn.Sequential(*[BuildingBlock(i, o) for i, o in [(3,32),(32,64),(64,128),(128,256)]]
            self.output = nn.Sequential(nn.Dropout(0.5), nn.Linear(2*2*256, 512), nn.ReLU(), nn.Linear(512, 100))

        def forward(self, x):
            return self.output(torch.flatten(self.conv(x), start_dim=1))

class CIFARWideModel(nn.Module):
    """Wider CNN: 3->64->128->256->512 channels, MLP 2048->512->100."""
    def __init__(self):
        super().__init__()
        self.conv = nn.Sequential(*[BuildingBlock(i, o) for i, o in [(3,64),(64,128),(128,256),(256,512)]]
        self.output = nn.Sequential(nn.Dropout(0.5), nn.Linear(2*2*512, 512), nn.ReLU(), nn.Linear(512, 100))

    def forward(self, x):
        return self.output(torch.flatten(self.conv(x), start_dim=1))

class CIFARDeepModelNoBN(nn.Module):
    """Deep CNN (wider channels, multiple convs per block), NO BatchNorm."""
    def __init__(self, num_convs_per_block=2):
        super().__init__()
        self.conv = nn.Sequential(*[DeepBlockNoBN(i, o, num_convs_per_block)
                                    for i, o in [(3,64),(64,128),(128,256),(256,512)]]
        self.output = nn.Sequential(nn.Dropout(0.5), nn.Linear(2*2*512, 512), nn.ReLU(), nn.Linear(512, 100))

    def forward(self, x):
        return self.output(torch.flatten(self.conv(x), start_dim=1))

class CIFARDeepModelBN(nn.Module):
    """Deep CNN (wider channels, multiple convs per block), WITH BatchNorm."""
    def __init__(self, num_convs_per_block=2):
        super().__init__()
        self.conv = nn.Sequential(*[DeepBlockBN(i, o, num_convs_per_block)
                                    for i, o in [(3,64),(64,128),(128,256),(256,512)]]
        self.output = nn.Sequential(nn.Dropout(0.5), nn.Linear(2*2*512, 512),
                                    nn.BatchNorm1d(512), nn.ReLU(), nn.Linear(512, 100))

    def forward(self, x):
        return self.output(torch.flatten(self.conv(x), start_dim=1))

```

Problem 1: Assessing the Original Fit

1.1 Load CIFAR100 (No Augmentation)

```
In [10]: transform_base = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize(cifar100_mean, cifar100_std)
])

cifar_train = datasets.CIFAR100(root="/content/data", train=True, download=True, transform=transform_base)
cifar_test = datasets.CIFAR100(root="/content/data", train=False, download=True, transform=transform_base)

cifar_dm = SimpleDataModule(cifar_train, cifar_test, validation=0.2, num_workers=2, batch_size=128)
print(f"Train: {len(cifar_train)}, Test: {len(cifar_test)}")
print(f"Classes: {len(cifar_train.classes)}")
```

100%|██████████| 169M/169M [00:13<00:00, 12.8MB/s]

Train: 50000, Test: 10000

Classes: 100

1.2 Preview a Batch

```
In [11]: data_loader = torch.utils.data.DataLoader(cifar_train, batch_size=10, shuffle=True)
imgs, lbls = next(iter(data_loader))

plt.figure(figsize=(15, 3))
grid = torchvision.utils.make_grid(nrow=10, tensor=imgs)
# Undo normalization for display
mean = torch.tensor(cifar100_mean).view(3,1,1)
std = torch.tensor(cifar100_std).view(3,1,1)
grid_show = torch.clamp(grid * std + mean, 0, 1)
plt.imshow(np.transpose(grid_show.numpy(), (1,2,0)))
plt.title("Sample CIFAR100 training images")
plt.axis("off")
plt.show()
print("Labels:", [cifar_train.classes[l] for l in lbls])
```

Sample CIFAR100 training images



Labels: ['crocodile', 'keyboard', 'cattle', 'orange', 'plate', 'house', 'ray', 'bicycle', 'apple', 'beaver']

1.3 Model Summary

```
In [12]: cifar_model = CIFARModel()
X_ = next(iter(cifar_dm.train_dataloader()))[0]
summary(cifar_model, input_data=X_, col_names=["input_size", "output_size", "num_params"])
```

```

Out[12]: =====
=====
Layer (type:depth-idx)      Input Shape      Output Shape      Param #
=====
CIFARModel                  [128, 3, 32, 32] [128, 100]        --
├─Sequential: 1-1           [128, 3, 32, 32] [128, 256, 2, 2]  --
│   └─BuildingBlock: 2-1    [128, 3, 32, 32] [128, 32, 16, 16] --
│       └─Conv2d: 3-1        [128, 3, 32, 32] [128, 32, 32, 32] 896
│           └─BatchNorm2d: 3-2 [128, 32, 32, 32] [128, 32, 32, 32] 64
│               └─ReLU: 3-3    [128, 32, 32, 32] [128, 32, 32, 32] --
│                   └─MaxPool2d: 3-4 [128, 32, 32, 32] [128, 32, 16, 16] --
│                       └─BuildingBlock: 2-2 [128, 32, 16, 16] [128, 64, 8, 8] --
│                           └─Conv2d: 3-5 [128, 32, 16, 16] [128, 64, 16, 16] 18,496
│                               └─BatchNorm2d: 3-6 [128, 64, 16, 16] [128, 64, 16, 16] 128
│                                   └─ReLU: 3-7 [128, 64, 16, 16] [128, 64, 16, 16] --
│                                       └─MaxPool2d: 3-8 [128, 64, 16, 16] [128, 64, 8, 8] --
│                                           └─BuildingBlock: 2-3 [128, 64, 8, 8] [128, 128, 4, 4] --
│                                               └─Conv2d: 3-9 [128, 64, 8, 8] [128, 128, 8, 8] 73,856
│                                                   └─BatchNorm2d: 3-10 [128, 128, 8, 8] [128, 128, 8, 8] 256
│                                                       └─ReLU: 3-11 [128, 128, 8, 8] [128, 128, 8, 8] --
│                                                           └─MaxPool2d: 3-12 [128, 128, 8, 8] [128, 128, 4, 4] --
│                                                               └─BuildingBlock: 2-4 [128, 128, 4, 4] [128, 256, 2, 2] --
│                                                                   └─Conv2d: 3-13 [128, 128, 4, 4] [128, 256, 4, 4] 295,168
│                                                                       └─BatchNorm2d: 3-14 [128, 256, 4, 4] [128, 256, 4, 4] 512
│                                                                           └─ReLU: 3-15 [128, 256, 4, 4] [128, 256, 4, 4] --
│                                                                               └─MaxPool2d: 3-16 [128, 256, 4, 4] [128, 256, 2, 2] --
├─Sequential: 1-2           [128, 1024] [128, 100]        --
│   └─Dropout: 2-5           [128, 1024] [128, 1024]        --
│       └─Linear: 2-6         [128, 1024] [128, 512]         524,800
│           └─ReLU: 2-7       [128, 512] [128, 512]        --
│               └─Linear: 2-8 [128, 512] [128, 100]         51,300
=====
Total params: 965,476
Trainable params: 965,476
Non-trainable params: 0
Total mult-adds (Units.GIGABYTES): 2.01
=====
Input size (MB): 1.57
Forward/backward pass size (MB): 126.46
Params size (MB): 3.86
Estimated Total Size (MB): 131.89
=====

```

1.4 Train for 25 Epochs

```

In [13]: cifar_optimizer = Adam(cifar_model.parameters(), lr=0.001, weight_decay=1e-4)
cifar_module = SimpleModule.classification(cifar_model, num_classes=100, optimizer=cifar_optimizer)
cifar_logger = CSVLogger(LOG_DIR, name="CIFAR100_P1")

checkpoint_p1 = ModelCheckpoint(monitor="valid_accuracy", mode="max", save_top_k=1,
                                filename="best-p1")

cifar_trainer = Trainer(
    deterministic=True, max_epochs=25, accelerator="gpu", devices=1,
    logger=cifar_logger,
    callbacks=[ErrorTracker(), TQDMProgressBar(refresh_rate=1), PrintMetrics(), checkpoint_p1]
)

cifar_trainer.fit(cifar_module, datamodule=cifar_dm)

```

```

INFO:pytorch_lightning.utilities.rank_zero:GPU available: True (cuda), used: True
INFO:pytorch_lightning.utilities.rank_zero:TPU available: False, using: 0 TPU cores
INFO:pytorch_lightning.utilities.rank_zero: Tip: For seamless cloud logging and experiment tracking,
try installing [litlogger](https://pypi.org/project/litlogger/) to enable LitLogger, which logs metric
s and artifacts automatically to the Lightning Experiments platform.
INFO:pytorch_lightning.accelerators.cuda:LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]

```

	Name	Type	Params	Mode	FLOPs
0	model	CIFARModel	965 K	train	0
1	loss	CrossEntropyLoss	0	train	0

Trainable params: 965 K

Non-trainable params: 0

Total params: 965 K

Total estimated model params size (MB): 3

Modules in train mode: 28

Modules in eval mode: 0

Total FLOPs: 0

Sanity Checking: | | 0/? [00:00<?, ?it/s]

/usr/local/lib/python3.12/dist-packages/pytorch_lightning/utilities/_pytree.py:21: `isininstance(treespec, LeafSpec)` is deprecated, use `isininstance(treespec, TreeSpec)` and `treespec.is\_leaf()` instead.

Training: | | 0/? [00:00<?, ?it/s]

Validation: | | 0/? [00:00<?, ?it/s]

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 1 | train_loss: 3.5472 | train_acc: 0.1539 | valid_loss: 2.7181 | valid_acc: 0.3058 | lr: 0.00
1000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 2 | train_loss: 2.7552 | train_acc: 0.2937 | valid_loss: 2.3317 | valid_acc: 0.3840 | lr: 0.00
1000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 3 | train_loss: 2.4243 | train_acc: 0.3612 | valid_loss: 2.2335 | valid_acc: 0.4086 | lr: 0.00
1000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 4 | train_loss: 2.2051 | train_acc: 0.4085 | valid_loss: 2.1945 | valid_acc: 0.4197 | lr: 0.00
1000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 5 | train_loss: 2.0278 | train_acc: 0.4466 | valid_loss: 2.0781 | valid_acc: 0.4461 | lr: 0.00
1000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 6 | train_loss: 1.8995 | train_acc: 0.4743 | valid_loss: 2.1114 | valid_acc: 0.4423 | lr: 0.00
1000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 7 | train_loss: 1.7751 | train_acc: 0.5038 | valid_loss: 2.0168 | valid_acc: 0.4661 | lr: 0.00
1000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 8 | train_loss: 1.6532 | train_acc: 0.5346 | valid_loss: 1.9478 | valid_acc: 0.4766 | lr: 0.00
1000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 9 | train_loss: 1.5632 | train_acc: 0.5531 | valid_loss: 1.9327 | valid_acc: 0.4848 | lr: 0.00
1000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 10 | train_loss: 1.4635 | train_acc: 0.5776 | valid_loss: 1.9078 | valid_acc: 0.5018 | lr: 0.00
1000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 11 | train_loss: 1.3768 | train_acc: 0.5983 | valid_loss: 1.9091 | valid_acc: 0.4947 | lr: 0.00
1000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 12 | train_loss: 1.2978 | train_acc: 0.6169 | valid_loss: 1.8797 | valid_acc: 0.5148 | lr: 0.00
1000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 13 | train_loss: 1.2158 | train_acc: 0.6403 | valid_loss: 1.9084 | valid_acc: 0.5056 | lr: 0.00
1000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 14 | train_loss: 1.1507 | train_acc: 0.6561 | valid_loss: 1.9947 | valid_acc: 0.5001 | lr: 0.00
1000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 15 | train_loss: 1.0946 | train_acc: 0.6700 | valid_loss: 1.9356 | valid_acc: 0.5120 | lr: 0.00
1000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 16 | train_loss: 1.0263 | train_acc: 0.6894 | valid_loss: 1.9519 | valid_acc: 0.5117 | lr: 0.00
1000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 17 | train_loss: 0.9723 | train_acc: 0.7035 | valid_loss: 2.0092 | valid_acc: 0.5056 | lr: 0.00
1000

```

Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 18 | train_loss: 0.9187 | train_acc: 0.7149 | valid_loss: 1.9667 | valid_acc: 0.5192 | lr: 0.00
1000
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 19 | train_loss: 0.8753 | train_acc: 0.7309 | valid_loss: 1.9969 | valid_acc: 0.5122 | lr: 0.00
1000
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 20 | train_loss: 0.8277 | train_acc: 0.7407 | valid_loss: 1.9996 | valid_acc: 0.5159 | lr: 0.00
1000
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 21 | train_loss: 0.7923 | train_acc: 0.7487 | valid_loss: 2.0208 | valid_acc: 0.5216 | lr: 0.00
1000
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 22 | train_loss: 0.7450 | train_acc: 0.7650 | valid_loss: 2.0253 | valid_acc: 0.5162 | lr: 0.00
1000
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 23 | train_loss: 0.7191 | train_acc: 0.7719 | valid_loss: 2.1163 | valid_acc: 0.5151 | lr: 0.00
1000
Validation: |           | 0/? [00:00<?, ?it/s]
INFO:pytorch_lightning.utilities.rank_zero:`Trainer.fit` stopped: `max_epochs=25` reached.
Epoch 24 | train_loss: 0.6901 | train_acc: 0.7797 | valid_loss: 2.0378 | valid_acc: 0.5239 | lr: 0.00
1000

```

1.5 Plot Training vs. Validation Accuracy

```

In [14]: epoch_metrics_p1 = read_epoch_metrics(cifar_logger)
print(epoch_metrics_p1[["train_loss", "train_accuracy_epoch", "valid_loss", "valid_accuracy"]])

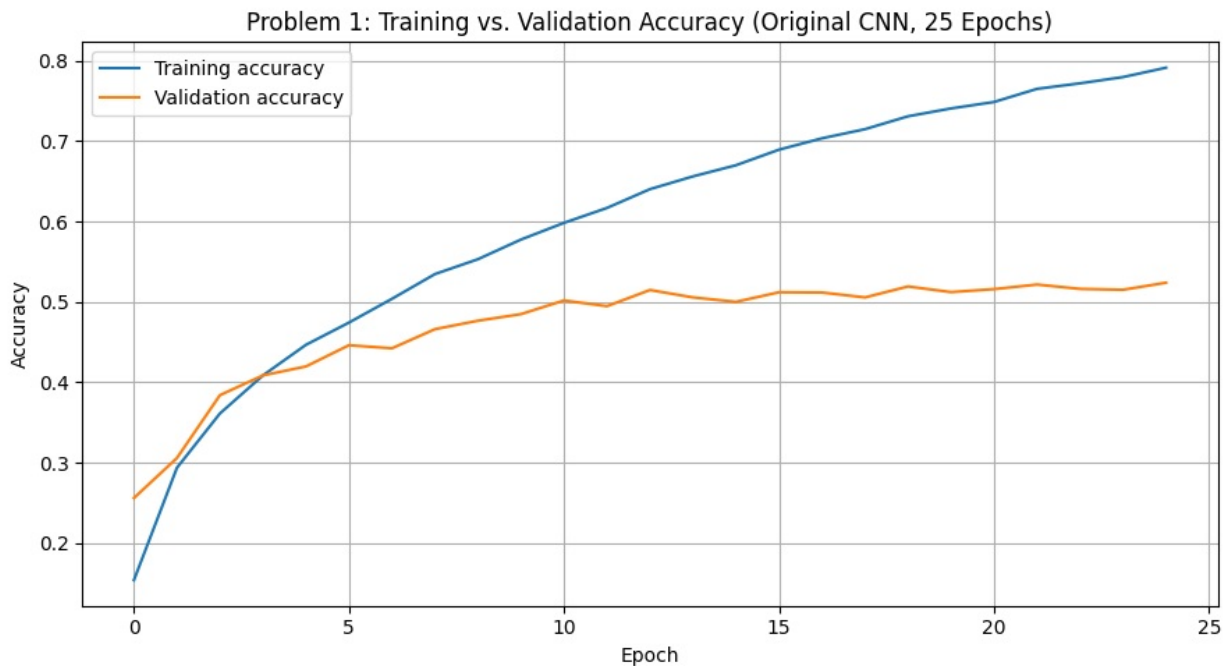
best_epoch_p1      = epoch_metrics_p1["valid_accuracy"].idxmax()
best_valid_acc_p1  = epoch_metrics_p1.loc[best_epoch_p1, "valid_accuracy"]
print(f"\nBest validation accuracy: {best_valid_acc_p1:.4f} at epoch {best_epoch_p1}")

plot_accuracy(epoch_metrics_p1, "Problem 1: Training vs. Validation Accuracy (Original CNN, 25 Epochs)

```

	train_loss	train_accuracy_epoch	valid_loss	valid_accuracy
epoch				
0	3.547197	0.153925	2.961274	0.2560
1	2.755168	0.293650	2.718097	0.3058
2	2.424299	0.361225	2.331679	0.3840
3	2.205060	0.408550	2.233520	0.4086
4	2.027829	0.446625	2.194510	0.4197
5	1.899494	0.474275	2.078080	0.4461
6	1.775082	0.503850	2.111383	0.4423
7	1.653171	0.534625	2.016786	0.4661
8	1.563157	0.553125	1.947807	0.4766
9	1.463460	0.577575	1.932691	0.4848
10	1.376767	0.598325	1.907800	0.5018
11	1.297794	0.616900	1.909141	0.4947
12	1.215799	0.640275	1.879707	0.5148
13	1.150726	0.656150	1.908446	0.5056
14	1.094576	0.670025	1.994694	0.5001
15	1.026272	0.689375	1.935552	0.5120
16	0.972331	0.703525	1.951935	0.5117
17	0.918696	0.714875	2.009248	0.5056
18	0.875263	0.730900	1.966686	0.5192
19	0.827656	0.740675	1.996868	0.5122
20	0.792275	0.748700	1.999577	0.5159
21	0.744954	0.765050	2.020821	0.5216
22	0.719102	0.771925	2.025301	0.5162
23	0.690128	0.779700	2.116305	0.5151
24	0.654926	0.791375	2.037807	0.5239

Best validation accuracy: 0.5239 at epoch 24



Problem 1(a) Interpretation

The accuracy curves show a clear overfitting pattern over the 25 training epochs. Training accuracy climbed steadily throughout, reaching around 0.79 by the final epoch, while validation accuracy plateaued much earlier — after roughly the first 10 epochs — and hovered between 0.51 and 0.52 for the remainder of training. The best validation accuracy was 0.5239, achieved at epoch 24, compared to a training accuracy of around 0.791 at the same point — a gap of over 26 percentage points. This divergence between the two curves is the signature of a model that is fitting training-specific patterns rather than learning features that generalize. The best checkpoint had a test accuracy of 0.5189 (test error: 0.4811).

1.6 Test-Set Accuracy (Saved for Final Table)

```
In [15]: test_results_p1 = cifar_trainer.test(cifar_module, datamodule=cifar_dm, ckpt_path="best", verbose=True)
best_model_test_accuracy = test_results_p1[0]["test_accuracy"]
print(f"Problem 1 test accuracy: {best_model_test_accuracy:.4f}")
print(f"Problem 1 test error: {1-best_model_test_accuracy:.4f}")
```

```
INFO:pytorch_lightning.utilities.rank_zero:Restoring states from the checkpoint path at /content/drive/MyDrive/hw7_logs/CIFAR100_P1/version_0/checkpoints/best-p1.ckpt
INFO:pytorch_lightning.accelerators.cuda:LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]
INFO:pytorch_lightning.utilities.rank_zero:Loaded model weights from the checkpoint at /content/drive/MyDrive/hw7_logs/CIFAR100_P1/version_0/checkpoints/best-p1.ckpt
/usr/local/lib/python3.12/dist-packages/pytorch_lightning/utilities/pytree.py:21: `isinstance(treespec, LeafSpec)` is deprecated, use `isinstance(treespec, TreeSpec)` and `treespec.is_leaf()` instead.
Testing: |          | 0/? [00:00<?, ?it/s]
Problem 1 test accuracy: 0.5189
Problem 1 test error: 0.4811
```

1.7 Per-Class Accuracy on Validation Set

```
In [16]: cifar_module.eval()
device = next(cifar_module.parameters()).device
```



```

all_preds, all_labels = [], []
with torch.no_grad():
    for x, y in cifar_dm.val_dataloader():
        preds = torch.argmax(cifar_module(x.to(device)), dim=1)
        all_preds.append(preds.cpu())
        all_labels.append(y)

all_preds = torch.cat(all_preds)
all_labels = torch.cat(all_labels)
class_names = cifar_train.classes

class_rows = []
for idx, name in enumerate(class_names):
    mask = all_labels == idx
    total = mask.sum().item()
    correct = (all_preds[mask] == all_labels[mask]).sum().item()
    class_rows.append({"class_index": idx, "class_name": name, "correct": correct,
                      "total": total, "accuracy": correct / total})

class_acc_df = pd.DataFrame(class_rows)

print("10 classes with HIGHEST validation accuracy:")
display(class_acc_df.nlargest(10, "accuracy"))

print("\n10 classes with LOWEST validation accuracy:")
display(class_acc_df.nsmallest(10, "accuracy"))

```

10 classes with HIGHEST validation accuracy:

	class_index	class_name	correct	total	accuracy
20	20	chair	98	112	0.875000
24	24	cockroach	95	111	0.855856
68	68	road	71	83	0.855422
75	75	skunk	91	107	0.850467
94	94	wardrobe	82	97	0.845361
69	69	rocket	80	99	0.808081
60	60	plain	67	83	0.807229
56	56	palm_tree	79	99	0.797980
0	0	apple	99	125	0.792000
48	48	motorcycle	64	81	0.790123

10 classes with LOWEST validation accuracy:

	class_index	class_name	correct	total	accuracy
72	72	seal	8	110	0.072727
50	50	mouse	13	83	0.156627
44	44	lizard	18	92	0.195652
35	35	girl	22	103	0.213592
93	93	turtle	25	103	0.242718
55	55	otter	26	96	0.270833
2	2	baby	29	105	0.276190
38	38	kangaroo	25	90	0.277778
4	4	beaver	28	98	0.285714
18	18	caterpillar	31	99	0.313131

Problem 1(b) Interpretation

There is substantial variation in accuracy across the 100 CIFAR100 classes. The ten best-performing categories — chair, cockroach, road, skunk, wardrobe, rocket, plain, palm_tree, apple, and motorcycle —

achieved validation accuracies ranging from about 0.79 to 0.875. These classes tend to have distinctive and consistent visual features: chair and wardrobe have recognizable geometric outlines, cockroach and skunk have strong color patterns, and road images are often dominated by uniform gray textures that are easy for the CNN to pick up at 32×32 resolution.

At the other end, the ten worst-performing classes — seal, mouse, lizard, girl, turtle, otter, baby, kangaroo, beaver, and caterpillar — all scored below 0.32, with seal as low as 0.073. Many of these are animal categories that look visually similar at low resolution: seal, otter, and beaver, for instance, share rounded body shapes and often appear against water or ground backgrounds that blur the distinctions between them. The girl and baby classes also present challenges due to highly variable poses, clothing, and backgrounds that the shallow CNN cannot reliably distinguish from other human or animal categories.

1.8 Visualize 5 Misclassified Images

```
In [17]: cifar_module.eval()
all_preds2, all_labels2, all_images2 = [], [], []

with torch.no_grad():
    for x, y in cifar_dm.val_dataloader():
        preds = torch.argmax(cifar_module(x.to(device)), dim=1)
        all_preds2.append(preds.cpu())
        all_labels2.append(y)
        all_images2.append(x.cpu())

all_preds2 = torch.cat(all_preds2)
all_labels2 = torch.cat(all_labels2)
all_images2 = torch.cat(all_images2)

wrong_mask = all_preds2 != all_labels2
wrong_images = all_images2[wrong_mask]
wrong_preds = all_preds2[wrong_mask]
wrong_labels = all_labels2[wrong_mask]

fig, axes = plt.subplots(1, 5, figsize=(15, 4))
for i, ax in enumerate(axes):
    img = unnormalize_cifar100(wrong_images[i]).permute(1, 2, 0)
    ax.imshow(img)
    ax.axis("off")
    ax.set_title(f"True: {class_names[wrong_labels[i]]}\nPred: {class_names[wrong_preds[i]]}", fontsi

plt.suptitle("Problem 1(c): Five Misclassified Validation Images", fontsize=13)
plt.tight_layout()
plt.show()
```



Problem 1(c) Interpretation

The five misclassified images span a range of error types. Some mistakes are easy to understand given the 32×32 resolution: a squirrel predicted as a seal, or a seal predicted as a rabbit, are cases where the animal's small size, rounded shape, and muted colors make the categories genuinely hard to separate. Similarly, a bottle predicted as a can is a reasonable confusion given that both are cylindrical objects that appear similar at low resolution.

The harder-to-explain case is a butterfly predicted as a beetle — while both are insects, butterflies tend to have very distinct wing patterns that should set them apart. This suggests the model may be picking up on background color or texture rather than the object itself in some cases, which is a known limitation of shallow CNNs trained on small images without strong regularization.

Problem 2: Augmenting the Data

2.1 Define Transforms

```
In [18]: # Augmented transform for training only
transform_train_aug = transforms.Compose([
    transforms.RandomHorizontalFlip(),
    transforms.RandomCrop(32, padding=4),
    transforms.ColorJitter(brightness=0.2, contrast=0.2, saturation=0.2),
    transforms.RandomRotation(5),
    transforms.ToTensor(),
    transforms.Normalize(cifar100_mean, cifar100_std),
    transforms.RandomErasing(p=0.5, scale=(0.02, 0.2))
])

# No augmentation for validation and test
transform_no_aug = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize(cifar100_mean, cifar100_std)
])
```

2.2 Build Augmented Datasets with Separate Val Transform

```
In [19]: full_train_aug = datasets.CIFAR100(root="/content/data", train=True, download=True, transform=transform_train_aug)
full_train_no_aug = datasets.CIFAR100(root="/content/data", train=True, download=True, transform=transform_no_aug)
cifar_test_aug = datasets.CIFAR100(root="/content/data", train=False, download=True, transform=transform_train_aug)

train_size = int(0.8 * len(full_train_aug))
valid_size = len(full_train_aug) - train_size

generator = torch.Generator().manual_seed(42)
train_split, valid_split = random_split(range(len(full_train_aug)), [train_size, valid_size], generator=generator)

cifar_train_aug = torch.utils.data.Subset(full_train_aug, train_split.indices)
cifar_valid_no_aug = torch.utils.data.Subset(full_train_no_aug, valid_split.indices)

cifar_dm_aug = SimpleDataModule(
    cifar_train_aug, cifar_test_aug,
    validation=cifar_valid_no_aug,
    num_workers=2, batch_size=256
)
print(f"Train: {len(cifar_train_aug)}, Val: {len(cifar_valid_no_aug)}, Test: {len(cifar_test_aug)}")
```

Train: 40000, Val: 10000, Test: 10000

2.3 Train Original Architecture with Augmented Data (75 Epochs)

```
In [20]: cifar_model_aug = CIFARModel()
cifar_optimizer_aug = Adam(cifar_model_aug.parameters(), lr=0.001, weight_decay=1e-4)
cifar_module_aug = SimpleModule.classification(cifar_model_aug, num_classes=100, optimizer=cifar_optimizer_aug)
cifar_logger_aug = CSVLogger(LOG_DIR, name="CIFAR100_P2_augmented")

checkpoint_p2 = ModelCheckpoint(monitor="valid_accuracy", mode="max", save_top_k=1, filename="best-p2")

cifar_trainer_aug = Trainer(
    deterministic=True, max_epochs=75, accelerator="gpu", devices=1,
    logger=cifar_logger_aug,
    callbacks=[ErrorTracker(), TQDMProgressBar(refresh_rate=1), PrintMetrics(), checkpoint_p2]
)

cifar_trainer_aug.fit(cifar_module_aug, datamodule=cifar_dm_aug)
```

```
INFO:pytorch_lightning.utilities.rank_zero:GPU available: True (cuda), used: True
INFO:pytorch_lightning.utilities.rank_zero:TPU available: False, using: 0 TPU cores
INFO:pytorch_lightning.utilities.rank_zero: Tip: For seamless cloud logging and experiment tracking,
try installing [litlogger](https://pypi.org/project/litlogger/) to enable LitLogger, which logs metric
s and artifacts automatically to the Lightning Experiments platform.
INFO:pytorch_lightning.accelerators.cuda:LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]
```

	Name	Type	Params	Mode	FLOPs
0	model	CIFARModel	965 K	train	0
1	loss	CrossEntropyLoss	0	train	0

Trainable params: 965 K

Non-trainable params: 0

Total params: 965 K

Total estimated model params size (MB): 3

Modules in train mode: 28

Modules in eval mode: 0

Total FLOPs: 0

Sanity Checking: | | 0/? [00:00<?, ?it/s]

/usr/local/lib/python3.12/dist-packages/pytorch_lightning/utilities/pytree.py:21: `isinstance(treespec, LeafSpec)` is deprecated, use `isinstance(treespec, TreeSpec)` and `treespec.is\_leaf()` instead.

Training: | | 0/? [00:00<?, ?it/s]

Validation: | | 0/? [00:00<?, ?it/s]

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 1 | train_loss: 3.9557 | train_acc: 0.0895 | valid_loss: 3.0239 | valid_acc: 0.2393 | lr: 0.001000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 2 | train_loss: 3.4305 | train_acc: 0.1695 | valid_loss: 2.8769 | valid_acc: 0.2636 | lr: 0.001000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 3 | train_loss: 3.1396 | train_acc: 0.2193 | valid_loss: 2.7082 | valid_acc: 0.3080 | lr: 0.001000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 4 | train_loss: 2.9407 | train_acc: 0.2579 | valid_loss: 2.6015 | valid_acc: 0.3300 | lr: 0.001000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 5 | train_loss: 2.8027 | train_acc: 0.2864 | valid_loss: 2.4073 | valid_acc: 0.3639 | lr: 0.001000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 6 | train_loss: 2.6932 | train_acc: 0.3076 | valid_loss: 2.4714 | valid_acc: 0.3634 | lr: 0.001000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 7 | train_loss: 2.5986 | train_acc: 0.3259 | valid_loss: 2.3296 | valid_acc: 0.3903 | lr: 0.001000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 8 | train_loss: 2.5225 | train_acc: 0.3417 | valid_loss: 2.2539 | valid_acc: 0.4052 | lr: 0.001000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 9 | train_loss: 2.4434 | train_acc: 0.3571 | valid_loss: 2.2131 | valid_acc: 0.4155 | lr: 0.001000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 10 | train_loss: 2.3895 | train_acc: 0.3701 | valid_loss: 2.1560 | valid_acc: 0.4219 | lr: 0.001000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 11 | train_loss: 2.3371 | train_acc: 0.3839 | valid_loss: 2.0941 | valid_acc: 0.4426 | lr: 0.001000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 12 | train_loss: 2.2913 | train_acc: 0.3941 | valid_loss: 2.0860 | valid_acc: 0.4404 | lr: 0.001000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 13 | train_loss: 2.2462 | train_acc: 0.4031 | valid_loss: 2.0810 | valid_acc: 0.4439 | lr: 0.001000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 14 | train_loss: 2.2096 | train_acc: 0.4102 | valid_loss: 2.0270 | valid_acc: 0.4572 | lr: 0.001000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 15 | train_loss: 2.1705 | train_acc: 0.4180 | valid_loss: 2.0498 | valid_acc: 0.4518 | lr: 0.001000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 16 | train_loss: 2.1345 | train_acc: 0.4268 | valid_loss: 1.9931 | valid_acc: 0.4633 | lr: 0.001000

Validation: | | 0/? [00:00<?, ?it/s]

Epoch 17 | train_loss: 2.1069 | train_acc: 0.4347 | valid_loss: 1.9255 | valid_acc: 0.4760 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 18 | train_loss: 2.0816 | train_acc: 0.4379 | valid_loss: 2.0648 | valid_acc: 0.4502 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 19 | train_loss: 2.0420 | train_acc: 0.4458 | valid_loss: 1.9034 | valid_acc: 0.4860 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 20 | train_loss: 2.0272 | train_acc: 0.4512 | valid_loss: 1.8856 | valid_acc: 0.4916 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 21 | train_loss: 1.9980 | train_acc: 0.4568 | valid_loss: 1.9381 | valid_acc: 0.4806 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 22 | train_loss: 1.9782 | train_acc: 0.4625 | valid_loss: 1.8773 | valid_acc: 0.4904 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 23 | train_loss: 1.9482 | train_acc: 0.4672 | valid_loss: 1.8361 | valid_acc: 0.5002 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 24 | train_loss: 1.9258 | train_acc: 0.4731 | valid_loss: 1.8793 | valid_acc: 0.4933 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 25 | train_loss: 1.9101 | train_acc: 0.4766 | valid_loss: 1.8436 | valid_acc: 0.4979 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 26 | train_loss: 1.8991 | train_acc: 0.4787 | valid_loss: 1.8642 | valid_acc: 0.4958 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 27 | train_loss: 1.8871 | train_acc: 0.4824 | valid_loss: 1.8525 | valid_acc: 0.4983 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 28 | train_loss: 1.8617 | train_acc: 0.4875 | valid_loss: 1.8156 | valid_acc: 0.5101 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 29 | train_loss: 1.8445 | train_acc: 0.4913 | valid_loss: 1.8276 | valid_acc: 0.5054 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 30 | train_loss: 1.8291 | train_acc: 0.4937 | valid_loss: 1.8037 | valid_acc: 0.5100 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 31 | train_loss: 1.8307 | train_acc: 0.4952 | valid_loss: 1.7695 | valid_acc: 0.5164 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 32 | train_loss: 1.8077 | train_acc: 0.5035 | valid_loss: 1.7704 | valid_acc: 0.5184 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 33 | train_loss: 1.7863 | train_acc: 0.5052 | valid_loss: 1.7615 | valid_acc: 0.5230 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 34 | train_loss: 1.7703 | train_acc: 0.5090 | valid_loss: 1.7538 | valid_acc: 0.5257 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 35 | train_loss: 1.7692 | train_acc: 0.5104 | valid_loss: 1.7769 | valid_acc: 0.5195 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 36 | train_loss: 1.7578 | train_acc: 0.5109 | valid_loss: 1.7723 | valid_acc: 0.5197 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 37 | train_loss: 1.7439 | train_acc: 0.5154 | valid_loss: 1.7283 | valid_acc: 0.5344 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 38 | train_loss: 1.7278 | train_acc: 0.5201 | valid_loss: 1.7470 | valid_acc: 0.5277 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 39 | train_loss: 1.7127 | train_acc: 0.5235 | valid_loss: 1.7226 | valid_acc: 0.5271 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 40 | train_loss: 1.7105 | train_acc: 0.5247 | valid_loss: 1.6759 | valid_acc: 0.5414 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 41 | train_loss: 1.7072 | train_acc: 0.5211 | valid_loss: 1.7426 | valid_acc: 0.5289 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]

Epoch 42 | train_loss: 1.6834 | train_acc: 0.5283 | valid_loss: 1.6855 | valid_acc: 0.5396 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 43 | train_loss: 1.6820 | train_acc: 0.5285 | valid_loss: 1.7382 | valid_acc: 0.5282 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 44 | train_loss: 1.6732 | train_acc: 0.5301 | valid_loss: 1.6925 | valid_acc: 0.5404 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 45 | train_loss: 1.6595 | train_acc: 0.5360 | valid_loss: 1.7046 | valid_acc: 0.5345 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 46 | train_loss: 1.6489 | train_acc: 0.5375 | valid_loss: 1.7394 | valid_acc: 0.5376 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 47 | train_loss: 1.6530 | train_acc: 0.5340 | valid_loss: 1.6802 | valid_acc: 0.5413 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 48 | train_loss: 1.6285 | train_acc: 0.5429 | valid_loss: 1.6862 | valid_acc: 0.5467 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 49 | train_loss: 1.6405 | train_acc: 0.5400 | valid_loss: 1.6746 | valid_acc: 0.5485 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 50 | train_loss: 1.6205 | train_acc: 0.5436 | valid_loss: 1.6656 | valid_acc: 0.5459 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 51 | train_loss: 1.6180 | train_acc: 0.5440 | valid_loss: 1.7305 | valid_acc: 0.5355 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 52 | train_loss: 1.6101 | train_acc: 0.5457 | valid_loss: 1.6914 | valid_acc: 0.5441 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 53 | train_loss: 1.5940 | train_acc: 0.5485 | valid_loss: 1.6666 | valid_acc: 0.5478 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 54 | train_loss: 1.5938 | train_acc: 0.5537 | valid_loss: 1.7077 | valid_acc: 0.5394 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 55 | train_loss: 1.5955 | train_acc: 0.5505 | valid_loss: 1.6572 | valid_acc: 0.5517 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 56 | train_loss: 1.5830 | train_acc: 0.5540 | valid_loss: 1.6683 | valid_acc: 0.5488 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 57 | train_loss: 1.5712 | train_acc: 0.5562 | valid_loss: 1.6775 | valid_acc: 0.5465 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 58 | train_loss: 1.5721 | train_acc: 0.5550 | valid_loss: 1.6629 | valid_acc: 0.5534 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 59 | train_loss: 1.5659 | train_acc: 0.5584 | valid_loss: 1.6596 | valid_acc: 0.5482 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 60 | train_loss: 1.5567 | train_acc: 0.5606 | valid_loss: 1.6476 | valid_acc: 0.5529 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 61 | train_loss: 1.5481 | train_acc: 0.5606 | valid_loss: 1.6524 | valid_acc: 0.5549 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 62 | train_loss: 1.5464 | train_acc: 0.5638 | valid_loss: 1.5953 | valid_acc: 0.5653 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 63 | train_loss: 1.5482 | train_acc: 0.5618 | valid_loss: 1.6057 | valid_acc: 0.5659 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 64 | train_loss: 1.5390 | train_acc: 0.5649 | valid_loss: 1.6488 | valid_acc: 0.5537 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 65 | train_loss: 1.5416 | train_acc: 0.5618 | valid_loss: 1.6010 | valid_acc: 0.5653 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 66 | train_loss: 1.5296 | train_acc: 0.5643 | valid_loss: 1.6342 | valid_acc: 0.5622 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]

```
Epoch 67 | train_loss: 1.5260 | train_acc: 0.5654 | valid_loss: 1.6258 | valid_acc: 0.5599 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 68 | train_loss: 1.5139 | train_acc: 0.5692 | valid_loss: 1.6604 | valid_acc: 0.5593 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 69 | train_loss: 1.5155 | train_acc: 0.5673 | valid_loss: 1.6009 | valid_acc: 0.5663 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 70 | train_loss: 1.5085 | train_acc: 0.5701 | valid_loss: 1.6328 | valid_acc: 0.5613 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 71 | train_loss: 1.4999 | train_acc: 0.5716 | valid_loss: 1.6054 | valid_acc: 0.5636 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 72 | train_loss: 1.5006 | train_acc: 0.5706 | valid_loss: 1.6156 | valid_acc: 0.5627 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 73 | train_loss: 1.4859 | train_acc: 0.5758 | valid_loss: 1.6039 | valid_acc: 0.5673 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
INFO:pytorch_lightning.utilities.rank_zero:`Trainer.fit` stopped: `max_epochs=75` reached.
Epoch 74 | train_loss: 1.4887 | train_acc: 0.5745 | valid_loss: 1.6241 | valid_acc: 0.5587 | lr: 0.00
1000
```

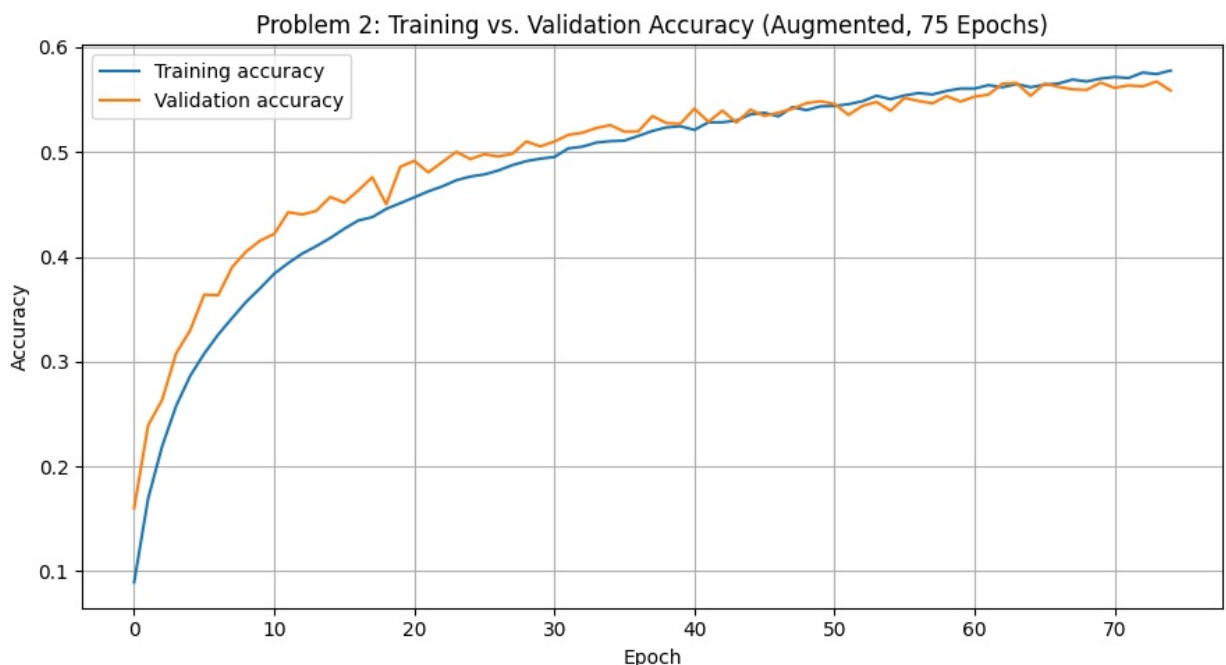
2.4 Plot and Report

```
In [21]: epoch_metrics_aug = read_epoch_metrics(cifar_logger_aug)
best_epoch_aug = epoch_metrics_aug["valid_accuracy"].idxmax()
best_valid_acc_aug = epoch_metrics_aug.loc[best_epoch_aug, "valid_accuracy"]
print(f"Best validation accuracy: {best_valid_acc_aug:.4f} at epoch {best_epoch_aug}")

plot_accuracy(epoch_metrics_aug, "Problem 2: Training vs. Validation Accuracy (Augmented, 75 Epochs)")

test_results_aug = cifar_trainer_aug.test(cifar_module_aug, datamodule=cifar_dm_aug, ckpt_path="be
problem2_test_accuracy = test_results_aug[0]["test_accuracy"]
print(f"Problem 2 test accuracy: {problem2_test_accuracy:.4f}")
print(f"Problem 2 test error: {1-problem2_test_accuracy:.4f}")
```

Best validation accuracy: 0.5673 at epoch 73



```
INFO:pytorch_lightning.utilities.rank_zero:Restoring states from the checkpoint path at /content/drive
/MyDrive/hw7_logs/CIFAR100_P2_augmented/version_0/checkpoints/best-p2.ckpt
INFO:pytorch_lightning.accelerators.cuda:LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]
INFO:pytorch_lightning.utilities.rank_zero:Loaded model weights from the checkpoint at /content/drive/
MyDrive/hw7_logs/CIFAR100_P2_augmented/version_0/checkpoints/best-p2.ckpt
/usr/local/lib/python3.12/dist-packages/pytorch_lightning/utilities/_pytree.py:21: `isinstance(treespec
c, LeafSpec)` is deprecated, use `isinstance(treespec, TreeSpec)` and `treespec.is_leaf()` instead.
Testing: | | 0/? [00:00<?, ?it/s]
```

Problem 2 test accuracy: 0.5754
Problem 2 test error: 0.4246

Problem 2 Interpretation

Retraining with data augmentation produced a noticeably different training dynamic compared to Problem 1. Applying random horizontal flips, random crops, color jitter, small rotations, and random erasing to the training data forces the model to learn features that are robust to these perturbations, rather than memorizing fixed pixel patterns.

The most visible effect is a much smaller gap between training and validation accuracy. While Problem 1 showed over a 26 percentage point divergence by epoch 25, the augmented model kept the two curves much closer together across all 75 epochs — a clear sign of reduced overfitting. The best validation accuracy improved to 0.5673 at epoch 73, compared to 0.5239 in Problem 1. The test accuracy of 0.5754 (test error: 0.4246) confirms the gain is real, representing an improvement of about 5.6 percentage points over Problem 1.

Problem 3: Widening the Network

3.1 Model Summary

```
In [22]: cifar_model_wide = CIFARWideModel()  
X_wide = next(iter(cifar_dm_aug.train_data_loader()))[0]  
summary(cifar_model_wide, input_data=X_wide, depth=3)
```

```
Out[22]: =====  
Layer (type:depth-idx)      Output Shape      Param #  
=====
```

CIFARWideModel	[256, 100]	--
└Sequential: 1-1	[256, 512, 2, 2]	--
└BuildingBlock: 2-1	[256, 64, 16, 16]	--
└Conv2d: 3-1	[256, 64, 32, 32]	1,792
└BatchNorm2d: 3-2	[256, 64, 32, 32]	128
└ReLU: 3-3	[256, 64, 32, 32]	--
└MaxPool2d: 3-4	[256, 64, 16, 16]	--
└BuildingBlock: 2-2	[256, 128, 8, 8]	--
└Conv2d: 3-5	[256, 128, 16, 16]	73,856
└BatchNorm2d: 3-6	[256, 128, 16, 16]	256
└ReLU: 3-7	[256, 128, 16, 16]	--
└MaxPool2d: 3-8	[256, 128, 8, 8]	--
└BuildingBlock: 2-3	[256, 256, 4, 4]	--
└Conv2d: 3-9	[256, 256, 8, 8]	295,168
└BatchNorm2d: 3-10	[256, 256, 8, 8]	512
└ReLU: 3-11	[256, 256, 8, 8]	--
└MaxPool2d: 3-12	[256, 256, 4, 4]	--
└BuildingBlock: 2-4	[256, 512, 2, 2]	--
└Conv2d: 3-13	[256, 512, 4, 4]	1,180,160
└BatchNorm2d: 3-14	[256, 512, 4, 4]	1,024
└ReLU: 3-15	[256, 512, 4, 4]	--
└MaxPool2d: 3-16	[256, 512, 2, 2]	--
└Sequential: 1-2	[256, 100]	--
└Dropout: 2-5	[256, 2048]	--
└Linear: 2-6	[256, 512]	1,049,088
└ReLU: 2-7	[256, 512]	--
└Linear: 2-8	[256, 100]	51,300

```
=====
```

Total params: 2,653,284
Trainable params: 2,653,284
Non-trainable params: 0
Total mult-adds (Units.GIGABYTES): 15.26

```
=====
```

Input size (MB): 3.15
Forward/backward pass size (MB): 504.57
Params size (MB): 10.61
Estimated Total Size (MB): 518.33

```
=====
```

3.2 Train the Wider Network (75 Epochs)


```
In [23]: cifar_optimizer_wide = Adam(cifar_model_wide.parameters(), lr=0.0005, weight_decay=1e-4)
cifar_module_wide = SimpleModule.classification(cifar_model_wide, num_classes=100, optimizer=cifar_optimizer_wide)
cifar_logger_wide = CSVLogger(LOG_DIR, name="CIFAR100_P3_wide")

checkpoint_p3 = ModelCheckpoint(monitor="valid_accuracy", mode="max", save_top_k=1, filename="best-p3")

cifar_trainer_wide = Trainer(
    deterministic=True, max_epochs=75, accelerator="gpu", devices=1,
    logger=cifar_logger_wide,
    callbacks=[ErrorTracker(), TQDMProgressBar(refresh_rate=1), PrintMetrics(), checkpoint_p3]
)

cifar_trainer_wide.fit(cifar_module_wide, datamodule=cifar_dm_aug)
```

```
INFO:pytorch_lightning.utilities.rank_zero:GPU available: True (cuda), used: True
INFO:pytorch_lightning.utilities.rank_zero:TPU available: False, using: 0 TPU cores
INFO:pytorch_lightning.utilities.rank_zero: Tip: For seamless cloud logging and experiment tracking,
try installing [litlogger](https://pypi.org/project/litlogger/) to enable LitLogger, which logs metric
s and artifacts automatically to the Lightning Experiments platform.
INFO:pytorch_lightning.accelerators.cuda:LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]
```

	Name	Type	Params	Mode	FLOPs
0	model	CIFARWideModel	2.7 M	train	0
1	loss	CrossEntropyLoss	0	train	0

Trainable params: 2.7 M

Non-trainable params: 0

Total params: 2.7 M

Total estimated model params size (MB): 10

Modules in train mode: 28

Modules in eval mode: 0

Total FLOPs: 0

```
Sanity Checking: |           | 0/? [00:00<?, ?it/s]
Training: |           | 0/? [00:00<?, ?it/s]
Validation: |           | 0/? [00:00<?, ?it/s]
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 1 | train_loss: 3.9214 | train_acc: 0.0986 | valid_loss: 2.9770 | valid_acc: 0.2621 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 2 | train_loss: 3.3144 | train_acc: 0.1894 | valid_loss: 2.6273 | valid_acc: 0.3280 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 3 | train_loss: 2.9883 | train_acc: 0.2497 | valid_loss: 2.6105 | valid_acc: 0.3359 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 4 | train_loss: 2.7872 | train_acc: 0.2896 | valid_loss: 2.3397 | valid_acc: 0.3816 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 5 | train_loss: 2.6435 | train_acc: 0.3203 | valid_loss: 2.2885 | valid_acc: 0.3947 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 6 | train_loss: 2.5065 | train_acc: 0.3494 | valid_loss: 2.2494 | valid_acc: 0.4067 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 7 | train_loss: 2.4172 | train_acc: 0.3681 | valid_loss: 2.2345 | valid_acc: 0.4144 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 8 | train_loss: 2.3137 | train_acc: 0.3900 | valid_loss: 2.1370 | valid_acc: 0.4307 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 9 | train_loss: 2.2561 | train_acc: 0.4020 | valid_loss: 2.0274 | valid_acc: 0.4610 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 10 | train_loss: 2.2011 | train_acc: 0.4165 | valid_loss: 2.0216 | valid_acc: 0.4591 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
```

```
Epoch 11 | train_loss: 2.1364 | train_acc: 0.4289 | valid_loss: 1.9872 | valid_acc: 0.4622 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 12 | train_loss: 2.0852 | train_acc: 0.4401 | valid_loss: 1.9329 | valid_acc: 0.4768 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 13 | train_loss: 2.0356 | train_acc: 0.4525 | valid_loss: 1.8778 | valid_acc: 0.4898 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 14 | train_loss: 1.9905 | train_acc: 0.4608 | valid_loss: 1.8603 | valid_acc: 0.4952 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 15 | train_loss: 1.9476 | train_acc: 0.4723 | valid_loss: 1.8469 | valid_acc: 0.5021 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 16 | train_loss: 1.9198 | train_acc: 0.4764 | valid_loss: 1.8418 | valid_acc: 0.4937 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 17 | train_loss: 1.8761 | train_acc: 0.4859 | valid_loss: 1.8515 | valid_acc: 0.5037 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 18 | train_loss: 1.8538 | train_acc: 0.4936 | valid_loss: 1.7721 | valid_acc: 0.5194 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 19 | train_loss: 1.8151 | train_acc: 0.5001 | valid_loss: 1.7608 | valid_acc: 0.5253 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 20 | train_loss: 1.7812 | train_acc: 0.5083 | valid_loss: 1.8414 | valid_acc: 0.5065 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 21 | train_loss: 1.7627 | train_acc: 0.5162 | valid_loss: 1.7205 | valid_acc: 0.5355 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 22 | train_loss: 1.7369 | train_acc: 0.5181 | valid_loss: 1.7421 | valid_acc: 0.5244 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 23 | train_loss: 1.7064 | train_acc: 0.5250 | valid_loss: 1.7690 | valid_acc: 0.5227 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 24 | train_loss: 1.6788 | train_acc: 0.5332 | valid_loss: 1.7107 | valid_acc: 0.5342 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 25 | train_loss: 1.6552 | train_acc: 0.5378 | valid_loss: 1.6963 | valid_acc: 0.5344 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 26 | train_loss: 1.6356 | train_acc: 0.5447 | valid_loss: 1.6729 | valid_acc: 0.5479 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 27 | train_loss: 1.6069 | train_acc: 0.5507 | valid_loss: 1.7040 | valid_acc: 0.5396 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 28 | train_loss: 1.5909 | train_acc: 0.5522 | valid_loss: 1.6387 | valid_acc: 0.5529 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 29 | train_loss: 1.5758 | train_acc: 0.5551 | valid_loss: 1.6211 | valid_acc: 0.5601 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 30 | train_loss: 1.5599 | train_acc: 0.5607 | valid_loss: 1.6852 | valid_acc: 0.5404 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 31 | train_loss: 1.5423 | train_acc: 0.5628 | valid_loss: 1.6431 | valid_acc: 0.5497 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 32 | train_loss: 1.5039 | train_acc: 0.5712 | valid_loss: 1.6506 | valid_acc: 0.5519 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 33 | train_loss: 1.5023 | train_acc: 0.5746 | valid_loss: 1.5876 | valid_acc: 0.5657 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 34 | train_loss: 1.4821 | train_acc: 0.5803 | valid_loss: 1.6110 | valid_acc: 0.5625 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 35 | train_loss: 1.4649 | train_acc: 0.5835 | valid_loss: 1.6263 | valid_acc: 0.5595 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
```

Epoch 36 | train_loss: 1.4486 | train_acc: 0.5874 | valid_loss: 1.5620 | valid_acc: 0.5722 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 37 | train_loss: 1.4381 | train_acc: 0.5889 | valid_loss: 1.5496 | valid_acc: 0.5799 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 38 | train_loss: 1.4221 | train_acc: 0.5943 | valid_loss: 1.5388 | valid_acc: 0.5783 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 39 | train_loss: 1.4083 | train_acc: 0.5976 | valid_loss: 1.6043 | valid_acc: 0.5678 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 40 | train_loss: 1.3884 | train_acc: 0.6019 | valid_loss: 1.5456 | valid_acc: 0.5791 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 41 | train_loss: 1.3754 | train_acc: 0.6051 | valid_loss: 1.5790 | valid_acc: 0.5709 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 42 | train_loss: 1.3555 | train_acc: 0.6103 | valid_loss: 1.5736 | valid_acc: 0.5781 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 43 | train_loss: 1.3445 | train_acc: 0.6142 | valid_loss: 1.5022 | valid_acc: 0.5904 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 44 | train_loss: 1.3399 | train_acc: 0.6127 | valid_loss: 1.5339 | valid_acc: 0.5864 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 45 | train_loss: 1.3204 | train_acc: 0.6201 | valid_loss: 1.5714 | valid_acc: 0.5810 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 46 | train_loss: 1.3151 | train_acc: 0.6190 | valid_loss: 1.6569 | valid_acc: 0.5701 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 47 | train_loss: 1.2917 | train_acc: 0.6279 | valid_loss: 1.5046 | valid_acc: 0.5885 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 48 | train_loss: 1.2866 | train_acc: 0.6265 | valid_loss: 1.5499 | valid_acc: 0.5852 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 49 | train_loss: 1.2702 | train_acc: 0.6342 | valid_loss: 1.4966 | valid_acc: 0.5995 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 50 | train_loss: 1.2596 | train_acc: 0.6334 | valid_loss: 1.5352 | valid_acc: 0.5962 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 51 | train_loss: 1.2499 | train_acc: 0.6373 | valid_loss: 1.5348 | valid_acc: 0.5898 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 52 | train_loss: 1.2520 | train_acc: 0.6359 | valid_loss: 1.5079 | valid_acc: 0.5959 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 53 | train_loss: 1.2301 | train_acc: 0.6431 | valid_loss: 1.5260 | valid_acc: 0.5934 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 54 | train_loss: 1.2291 | train_acc: 0.6401 | valid_loss: 1.5112 | valid_acc: 0.5956 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 55 | train_loss: 1.2152 | train_acc: 0.6465 | valid_loss: 1.5344 | valid_acc: 0.5978 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 56 | train_loss: 1.1941 | train_acc: 0.6495 | valid_loss: 1.4970 | valid_acc: 0.6007 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 57 | train_loss: 1.1897 | train_acc: 0.6481 | valid_loss: 1.4905 | valid_acc: 0.5973 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 58 | train_loss: 1.1909 | train_acc: 0.6526 | valid_loss: 1.5184 | valid_acc: 0.5933 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 59 | train_loss: 1.1685 | train_acc: 0.6558 | valid_loss: 1.4797 | valid_acc: 0.6081 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 60 | train_loss: 1.1629 | train_acc: 0.6601 | valid_loss: 1.5899 | valid_acc: 0.5981 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]

```

Epoch 61 | train_loss: 1.1555 | train_acc: 0.6597 | valid_loss: 1.5538 | valid_acc: 0.5957 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 62 | train_loss: 1.1356 | train_acc: 0.6647 | valid_loss: 1.4990 | valid_acc: 0.6014 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 63 | train_loss: 1.1476 | train_acc: 0.6606 | valid_loss: 1.4917 | valid_acc: 0.6044 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 64 | train_loss: 1.1389 | train_acc: 0.6648 | valid_loss: 1.5056 | valid_acc: 0.6024 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 65 | train_loss: 1.1191 | train_acc: 0.6692 | valid_loss: 1.5337 | valid_acc: 0.5925 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 66 | train_loss: 1.1071 | train_acc: 0.6719 | valid_loss: 1.5185 | valid_acc: 0.6006 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 67 | train_loss: 1.1009 | train_acc: 0.6758 | valid_loss: 1.4978 | valid_acc: 0.6059 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 68 | train_loss: 1.0973 | train_acc: 0.6742 | valid_loss: 1.5169 | valid_acc: 0.6023 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 69 | train_loss: 1.0866 | train_acc: 0.6771 | valid_loss: 1.4934 | valid_acc: 0.6124 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 70 | train_loss: 1.0796 | train_acc: 0.6809 | valid_loss: 1.5045 | valid_acc: 0.6078 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 71 | train_loss: 1.0764 | train_acc: 0.6781 | valid_loss: 1.4634 | valid_acc: 0.6124 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 72 | train_loss: 1.0731 | train_acc: 0.6826 | valid_loss: 1.4897 | valid_acc: 0.6077 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 73 | train_loss: 1.0654 | train_acc: 0.6809 | valid_loss: 1.5023 | valid_acc: 0.6101 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
INFO:pytorch_lightning.utilities.rank_zero:`Trainer.fit` stopped: `max_epochs=75` reached.
Epoch 74 | train_loss: 1.0482 | train_acc: 0.6888 | valid_loss: 1.5690 | valid_acc: 0.5968 | lr: 0.00
0500

```

3.3 Plot and Report

```

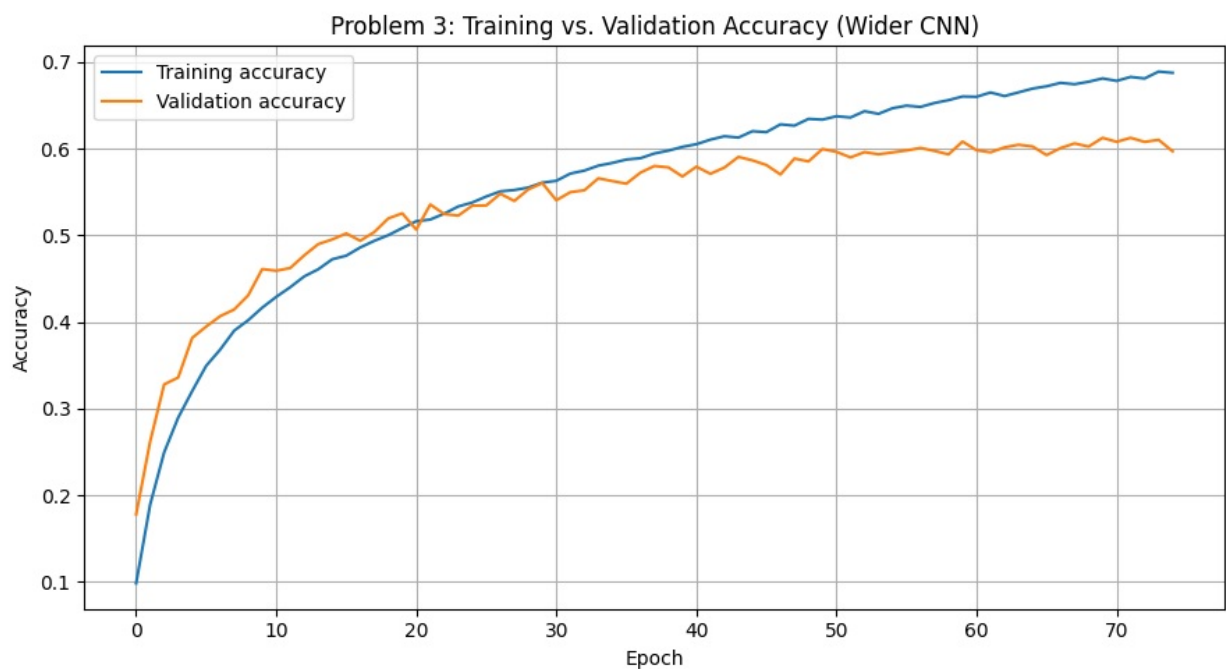
In [24]: epoch_metrics_wide = read_epoch_metrics(cifar_logger_wide)
best_epoch_wide = epoch_metrics_wide["valid_accuracy"].idxmax()
best_valid_acc_wide = epoch_metrics_wide.loc[best_epoch_wide, "valid_accuracy"]
print(f"Best validation accuracy: {best_valid_acc_wide:.4f} at epoch {best_epoch_wide}")

plot_accuracy(epoch_metrics_wide, "Problem 3: Training vs. Validation Accuracy (Wider CNN)")

test_results_wide = cifar_trainer_wide.test(cifar_module_wide, datamodule=cifar_dm_aug, ckpt_path=best_ckpt_path)
problem3_test_accuracy = test_results_wide[0]["test_accuracy"]
print(f"Problem 3 test accuracy: {problem3_test_accuracy:.4f}")
print(f"Problem 3 test error: {1-problem3_test_accuracy:.4f}")

```

Best validation accuracy: 0.6124 at epoch 69



```
INFO:pytorch_lightning.utilities.rank_zero:Restoring states from the checkpoint path at /content/drive/MyDrive/hw7_logs/CIFAR100_P3_wide/version_0/checkpoints/best-p3.ckpt
INFO:pytorch_lightning.accelerators.cuda:LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]
INFO:pytorch_lightning.utilities.rank_zero:Loaded model weights from the checkpoint at /content/drive/MyDrive/hw7_logs/CIFAR100_P3_wide/version_0/checkpoints/best-p3.ckpt
/usr/local/lib/python3.12/dist-packages/pytorch_lightning/utilities/_pytree.py:21: `isinstance(treespec, LeafSpec)` is deprecated, use `isinstance(treespec, TreeSpec)` and `treespec.is_leaf()` instead.
Testing: |          | 0/? [00:00<?, ?it/s]
Problem 3 test accuracy: 0.6134
Problem 3 test error:    0.3866
```

Problem 3 Interpretation

Widening the convolutional layers — doubling the number of channels at each block from 32/64/128/256 to 64/128/256/512 — gave the model substantially more capacity to detect diverse visual features simultaneously. With more filters at each stage, the network can track a larger number of patterns at once, from low-level edges and color gradients in the early layers to more complex shapes and textures in the deeper ones.

This paid off clearly: using the same augmented data as Problem 2 but with the wider architecture, the best validation accuracy reached 0.6124 at epoch 69, and the test accuracy improved to 0.6134 (test error: 0.3866) — a gain of roughly 3.8 percentage points over Problem 2. The linear classifier head was updated to accept `2 * 2 * 512 = 2048` inputs to match the wider feature maps from the final convolutional block.

Problem 4: Deep Learning

4(a): Deeper Network WITHOUT BatchNorm

4a.1 Model Summary

```
In [25]: cifar_model_deep = CIFARDeepModelNoBN(num_convs_per_block=2)
X_deep = next(iter(cifar_dm_aug.train_data_loader()))[0]
summary(cifar_model_deep, input_data=X_deep, depth=4)
```

```
Out[25]: =====
Layer (type:depth-idx)                               Output Shape                               Param #
=====
CIFARDeepModelNoBN                                   [256, 100]                                --
├─Sequential: 1-1                                     [256, 512, 2, 2]                           --
│   └─DeepBlockNoBN: 2-1                               [256, 64, 16, 16]                           --
│       └─Sequential: 3-1                               [256, 64, 16, 16]                           --
│           └─Conv2d: 4-1                               [256, 64, 32, 32]                           1,792
│               └─ReLU: 4-2                             [256, 64, 32, 32]                           --
│                   └─Conv2d: 4-3                       [256, 64, 32, 32]                           36,928
│                       └─ReLU: 4-4                     [256, 64, 32, 32]                           --
│                           └─MaxPool2d: 4-5            [256, 64, 16, 16]                           --
│                               └─DeepBlockNoBN: 2-2     [256, 128, 8, 8]                            --
│                                   └─Sequential: 3-2     [256, 128, 8, 8]                            --
│                                       └─Conv2d: 4-6     [256, 128, 16, 16]                          73,856
│                                           └─ReLU: 4-7    [256, 128, 16, 16]                          --
│                                               └─Conv2d: 4-8 [256, 128, 16, 16]                          147,584
│                                                   └─ReLU: 4-9 [256, 128, 16, 16]                          --
│                                                       └─MaxPool2d: 4-10 [256, 128, 8, 8]                            --
│                                                           └─DeepBlockNoBN: 2-3 [256, 256, 4, 4]                            --
│                                                               └─Sequential: 3-3 [256, 256, 4, 4]                            --
│                                                                   └─Conv2d: 4-11 [256, 256, 8, 8]                          295,168
│                                                                       └─ReLU: 4-12 [256, 256, 8, 8]                          --
│                                                                           └─Conv2d: 4-13 [256, 256, 8, 8]                          590,080
│                                                                               └─ReLU: 4-14 [256, 256, 8, 8]                          --
│                                                                                   └─MaxPool2d: 4-15 [256, 256, 4, 4]                            --
│                                                                                       └─DeepBlockNoBN: 2-4 [256, 512, 2, 2]                            --
│                                                                                           └─Sequential: 3-4 [256, 512, 2, 2]                            --
│                                                                                               └─Conv2d: 4-16 [256, 512, 4, 4]                          1,180,160
│                                                                                                   └─ReLU: 4-17 [256, 512, 4, 4]                          --
│                                                                                                       └─Conv2d: 4-18 [256, 512, 4, 4]                          2,359,808
│                                                                                                           └─ReLU: 4-19 [256, 512, 4, 4]                          --
│                                                                                                               └─MaxPool2d: 4-20 [256, 512, 2, 2]                            --
└─Sequential: 1-2                                     [256, 100]                                --
    └─Dropout: 2-5                                     [256, 2048]                               --
        └─Linear: 2-6                                   [256, 512]                                1,049,088
            └─ReLU: 2-7                                 [256, 512]                                --
                └─Linear: 2-8                           [256, 100]                                51,300
=====

Total params: 5,785,764
Trainable params: 5,785,764
Non-trainable params: 0
Total mult-adds (Units.GIGABYTES): 53.95
=====

Input size (MB): 3.15
Forward/backward pass size (MB): 504.57
Params size (MB): 23.14
Estimated Total Size (MB): 530.86
=====
```

4a.2 Train (40 Epochs)

```
In [26]: cifar_optimizer_deep = Adam(cifar_model_deep.parameters(), lr=0.0005, weight_decay=1e-4)
cifar_module_deep = SimpleModule.classification(cifar_model_deep, num_classes=100, optimizer=cifar_optimizer_deep)
cifar_logger_deep = CSVLogger(LOG_DIR, name="CIFAR100_P4a_deep_no_bn")

checkpoint_p4a = ModelCheckpoint(monitor="valid_accuracy", mode="max", save_top_k=1, filename="best-p4a")

cifar_trainer_deep = Trainer(
    deterministic=True, max_epochs=40, accelerator="gpu", devices=1,
    logger=cifar_logger_deep,
    callbacks=[ErrorTracker(), TQDMProgressBar(refresh_rate=1), PrintMetrics(), checkpoint_p4a]
)

cifar_trainer_deep.fit(cifar_module_deep, datamodule=cifar_dm_aug)
```

INFO:pytorch_lightning.utilities.rank_zero:GPU available: True (cuda), used: True
 INFO:pytorch_lightning.utilities.rank_zero:TPU available: False, using: 0 TPU cores
 INFO:pytorch_lightning.utilities.rank_zero: Tip: For seamless cloud logging and experiment tracking, try installing [litlogger](https://pypi.org/project/litlogger/) to enable LitLogger, which logs metrics and artifacts automatically to the Lightning Experiments platform.
 INFO:pytorch_lightning.accelerators.cuda:LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]

	Name	Type	Params	Mode	FLOPs
0	model	CIFARDeepModelNoBN	5.8 M	train	0
1	loss	CrossEntropyLoss	0	train	0

Trainable params: 5.8 M

Non-trainable params: 0

Total params: 5.8 M

Total estimated model params size (MB): 23

Modules in train mode: 36

Modules in eval mode: 0

Total FLOPs: 0

```

Sanity Checking: |           | 0/? [00:00<?, ?it/s]
Training: |           | 0/? [00:00<?, ?it/s]
Validation: |           | 0/? [00:00<?, ?it/s]
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 1 | train_loss: 4.4697 | train_acc: 0.0191 | valid_loss: 3.9768 | valid_acc: 0.0630 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 2 | train_loss: 4.1886 | train_acc: 0.0445 | valid_loss: 3.8072 | valid_acc: 0.0987 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 3 | train_loss: 3.9707 | train_acc: 0.0711 | valid_loss: 3.6456 | valid_acc: 0.1275 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 4 | train_loss: 3.8052 | train_acc: 0.1006 | valid_loss: 3.4565 | valid_acc: 0.1581 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 5 | train_loss: 3.6699 | train_acc: 0.1225 | valid_loss: 3.3098 | valid_acc: 0.1892 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 6 | train_loss: 3.5358 | train_acc: 0.1464 | valid_loss: 3.2845 | valid_acc: 0.1916 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 7 | train_loss: 3.4260 | train_acc: 0.1676 | valid_loss: 3.1554 | valid_acc: 0.2167 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 8 | train_loss: 3.3292 | train_acc: 0.1861 | valid_loss: 3.1573 | valid_acc: 0.2165 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 9 | train_loss: 3.2267 | train_acc: 0.2028 | valid_loss: 2.9658 | valid_acc: 0.2587 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 10 | train_loss: 3.1428 | train_acc: 0.2177 | valid_loss: 2.8494 | valid_acc: 0.2724 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 11 | train_loss: 3.0725 | train_acc: 0.2320 | valid_loss: 2.8128 | valid_acc: 0.2779 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 12 | train_loss: 2.9890 | train_acc: 0.2467 | valid_loss: 2.7174 | valid_acc: 0.2970 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 13 | train_loss: 2.9059 | train_acc: 0.2625 | valid_loss: 2.7365 | valid_acc: 0.2992 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 14 | train_loss: 2.8211 | train_acc: 0.2760 | valid_loss: 2.5842 | valid_acc: 0.3238 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 15 | train_loss: 2.7436 | train_acc: 0.2923 | valid_loss: 2.5469 | valid_acc: 0.3356 | lr: 0.00
0500
Validation: |           | 0/? [00:00<?, ?it/s]

```

```

Epoch 16 | train_loss: 2.6728 | train_acc: 0.3075 | valid_loss: 2.4833 | valid_acc: 0.3437 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 17 | train_loss: 2.6044 | train_acc: 0.3189 | valid_loss: 2.4060 | valid_acc: 0.3659 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 18 | train_loss: 2.5545 | train_acc: 0.3305 | valid_loss: 2.4249 | valid_acc: 0.3651 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 19 | train_loss: 2.4904 | train_acc: 0.3447 | valid_loss: 2.3061 | valid_acc: 0.3887 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 20 | train_loss: 2.4577 | train_acc: 0.3527 | valid_loss: 2.2959 | valid_acc: 0.3905 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 21 | train_loss: 2.3920 | train_acc: 0.3624 | valid_loss: 2.2446 | valid_acc: 0.4020 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 22 | train_loss: 2.3345 | train_acc: 0.3787 | valid_loss: 2.2374 | valid_acc: 0.4073 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 23 | train_loss: 2.2964 | train_acc: 0.3865 | valid_loss: 2.1957 | valid_acc: 0.4191 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 24 | train_loss: 2.2615 | train_acc: 0.3914 | valid_loss: 2.1197 | valid_acc: 0.4253 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 25 | train_loss: 2.2105 | train_acc: 0.4026 | valid_loss: 2.1042 | valid_acc: 0.4370 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 26 | train_loss: 2.1878 | train_acc: 0.4094 | valid_loss: 2.1548 | valid_acc: 0.4223 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 27 | train_loss: 2.1400 | train_acc: 0.4178 | valid_loss: 2.0742 | valid_acc: 0.4417 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 28 | train_loss: 2.1298 | train_acc: 0.4227 | valid_loss: 2.0347 | valid_acc: 0.4511 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 29 | train_loss: 2.0871 | train_acc: 0.4331 | valid_loss: 2.0130 | valid_acc: 0.4563 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 30 | train_loss: 2.0380 | train_acc: 0.4407 | valid_loss: 2.0321 | valid_acc: 0.4493 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 31 | train_loss: 2.0135 | train_acc: 0.4485 | valid_loss: 1.9656 | valid_acc: 0.4653 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 32 | train_loss: 1.9817 | train_acc: 0.4533 | valid_loss: 2.0410 | valid_acc: 0.4553 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 33 | train_loss: 1.9585 | train_acc: 0.4600 | valid_loss: 1.9780 | valid_acc: 0.4689 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 34 | train_loss: 1.9361 | train_acc: 0.4663 | valid_loss: 1.9451 | valid_acc: 0.4717 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 35 | train_loss: 1.9247 | train_acc: 0.4685 | valid_loss: 1.9432 | valid_acc: 0.4789 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 36 | train_loss: 1.8803 | train_acc: 0.4775 | valid_loss: 1.9248 | valid_acc: 0.4792 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 37 | train_loss: 1.8659 | train_acc: 0.4840 | valid_loss: 1.8944 | valid_acc: 0.4863 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 38 | train_loss: 1.8305 | train_acc: 0.4874 | valid_loss: 1.8801 | valid_acc: 0.4918 | lr: 0.00
0500
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 39 | train_loss: 1.8203 | train_acc: 0.4934 | valid_loss: 1.8439 | valid_acc: 0.4927 | lr: 0.00
0500
INFO:pytorch_lightning.utilities.rank_zero:`Trainer.fit` stopped: `max_epochs=40` reached.

```

4a.3 Plot and Report

```
In [27]: epoch_metrics_deep = read_epoch_metrics(cifar_logger_deep)
```



```

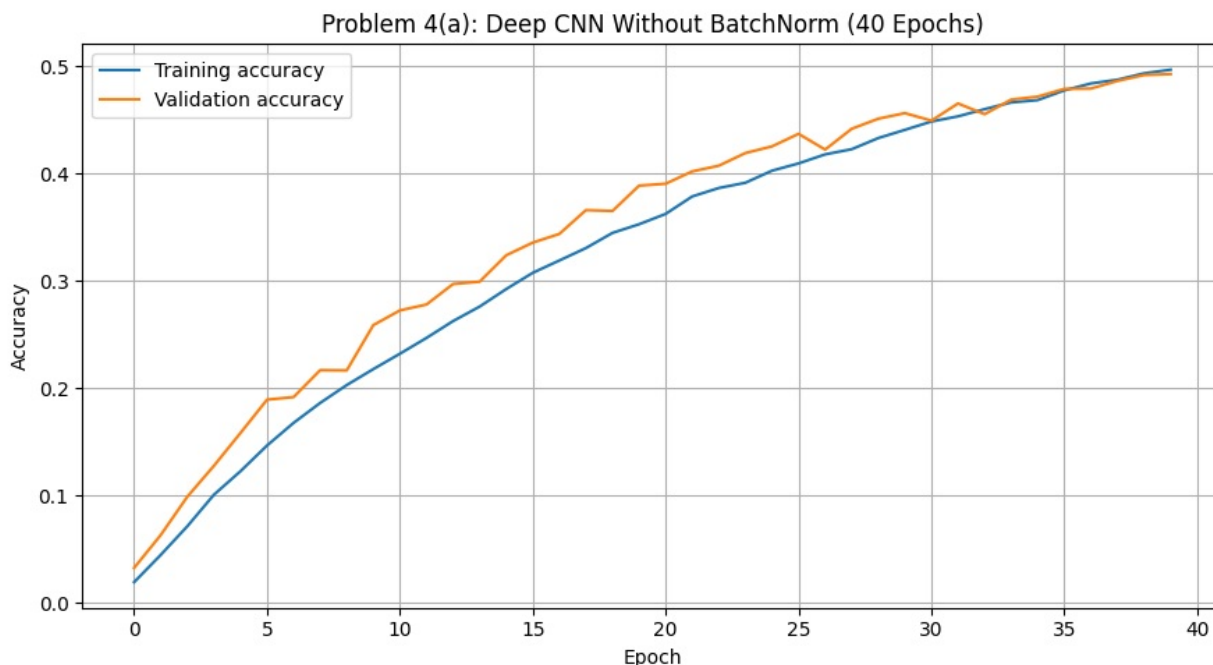
best_epoch_deep = epoch_metrics_deep["valid_accuracy"].idxmax()
best_valid_acc_deep = epoch_metrics_deep.loc[best_epoch_deep, "valid_accuracy"]
print(f"Best validation accuracy: {best_valid_acc_deep:.4f} at epoch {best_epoch_deep}")

plot_accuracy(epoch_metrics_deep, "Problem 4(a): Deep CNN Without BatchNorm (40 Epochs)")

test_results_deep = cifar_trainer_deep.test(cifar_module_deep, datamodule=cifar_dm_aug, ckpt_pa
problem4a_test_accuracy = test_results_deep[0]["test_accuracy"]
print(f"Problem 4(a) test accuracy: {problem4a_test_accuracy:.4f}")
print(f"Problem 4(a) test error: {1-problem4a_test_accuracy:.4f}")

```

Best validation accuracy: 0.4927 at epoch 39



```

INFO:pytorch_lightning.utilities.rank_zero:Restoring states from the checkpoint path at /content/drive/
/MyDrive/hw7_logs/CIFAR100_P4a_deep_no_bn/version_0/checkpoints/best-p4a.ckpt
INFO:pytorch_lightning.accelerators.cuda:LOCAL RANK: 0 - CUDA_VISIBLE_DEVICES: [0]
INFO:pytorch_lightning.utilities.rank_zero:Loaded model weights from the checkpoint at /content/drive/
MyDrive/hw7_logs/CIFAR100_P4a_deep_no_bn/version_0/checkpoints/best-p4a.ckpt
/usr/local/lib/python3.12/dist-packages/pytorch_lightning/utilities/_pytree.py:21: `isinstance(treespec
c, LeafSpec)` is deprecated, use `isinstance(treespec, TreeSpec)` and `treespec.is_leaf()` instead.
Testing: |          | 0/? [00:00<?, ?it/s]
Problem 4(a) test accuracy: 0.4988
Problem 4(a) test error:    0.5012

```

Problem 4(a) Interpretation

Adding a second convolutional layer to each block — without batch normalization — made training noticeably harder than the single-conv architecture. The model did learn, reaching a best validation accuracy of 0.4927 at epoch 39 and a test accuracy of 0.4988, but this is significantly worse than the shallower wide model in Problem 3 (0.6134), and the training accuracy was still pulling well ahead of validation by the end of the 40 epochs.

The deeper stack of nonlinear transformations without any normalization makes it harder for gradients to flow back cleanly to the early layers, slowing learning and making the model more susceptible to the vanishing gradient and dead neuron problems. While the network did not completely fail to learn as might happen with even more layers, the degraded performance relative to Problem 3 demonstrates that simply stacking more convolutional layers is not sufficient — the depth increase needs to be paired with normalization to be effective, as demonstrated in Problem 4(b).

4(b): Deeper Network WITH BatchNorm

4b.1 Model Summary

```
In [28]: cifar_model_deep_bn = CIFARDeepModelBN(num_convs_per_block=2)
X_deep_bn = next(iter(cifar_dm_aug.train_data_loader()))[0]
summary(cifar_model_deep_bn, input_data=X_deep_bn, depth=4)
```

```
Out[28]: =====
Layer (type:depth-idx)          Output Shape          Param #
=====
CIFARDeepModelBN                [256, 100]            --
├─Sequential: 1-1                [256, 512, 2, 2]      --
│   └─DeepBlockBN: 2-1            [256, 64, 16, 16]      --
│       └─Sequential: 3-1          [256, 64, 16, 16]      --
│           └─Conv2d: 4-1          [256, 64, 32, 32]      1,792
│               └─BatchNorm2d: 4-2 [256, 64, 32, 32]      128
│                   └─ReLU: 4-3    [256, 64, 32, 32]      --
│                       └─Conv2d: 4-4 [256, 64, 32, 32]      36,928
│                           └─BatchNorm2d: 4-5 [256, 64, 32, 32]      128
│                               └─ReLU: 4-6    [256, 64, 32, 32]      --
│                                   └─MaxPool2d: 4-7 [256, 64, 16, 16]      --
│                                       └─DeepBlockBN: 2-2            [256, 128, 8, 8]      --
│                                           └─Sequential: 3-2          [256, 128, 8, 8]      --
│                                               └─Conv2d: 4-8          [256, 128, 16, 16]      73,856
│                                                   └─BatchNorm2d: 4-9 [256, 128, 16, 16]      256
│                                                       └─ReLU: 4-10    [256, 128, 16, 16]      --
│                                                           └─Conv2d: 4-11 [256, 128, 16, 16]      147,584
│                                                               └─BatchNorm2d: 4-12 [256, 128, 16, 16]      256
│                                                                   └─ReLU: 4-13    [256, 128, 16, 16]      --
│                                                                       └─MaxPool2d: 4-14 [256, 128, 8, 8]      --
│                                                                           └─DeepBlockBN: 2-3            [256, 256, 4, 4]      --
│                                                                               └─Sequential: 3-3          [256, 256, 4, 4]      --
│                                                                                   └─Conv2d: 4-15          [256, 256, 8, 8]      295,168
│                                                                                       └─BatchNorm2d: 4-16 [256, 256, 8, 8]      512
│                                                                                           └─ReLU: 4-17    [256, 256, 8, 8]      --
│                                                                                               └─Conv2d: 4-18 [256, 256, 8, 8]      590,080
│                                                                                                   └─BatchNorm2d: 4-19 [256, 256, 8, 8]      512
│                                                                                                       └─ReLU: 4-20    [256, 256, 8, 8]      --
│                                                                                       └─MaxPool2d: 4-21 [256, 256, 4, 4]      --
│                                                                                           └─DeepBlockBN: 2-4            [256, 512, 2, 2]      --
│                                                                                               └─Sequential: 3-4          [256, 512, 2, 2]      --
│                                                                                                   └─Conv2d: 4-22          [256, 512, 4, 4]      1,180,160
│                                                                                                       └─BatchNorm2d: 4-23 [256, 512, 4, 4]      1,024
│                                                                                                           └─ReLU: 4-24    [256, 512, 4, 4]      --
│                                                                                                               └─Conv2d: 4-25          [256, 512, 4, 4]      2,359,808
│                                                                                               └─BatchNorm2d: 4-26 [256, 512, 4, 4]      1,024
│                                                                                                   └─ReLU: 4-27    [256, 512, 4, 4]      --
│                                                                                                       └─MaxPool2d: 4-28 [256, 512, 2, 2]      --
└─Sequential: 1-2                [256, 100]            --
    └─Dropout: 2-5                [256, 2048]           --
        └─Linear: 2-6              [256, 512]            1,049,088
            └─BatchNorm1d: 2-7     [256, 512]            1,024
                └─ReLU: 2-8        [256, 512]            --
                    └─Linear: 2-9   [256, 100]            51,300
=====

Total params: 5,790,628
Trainable params: 5,790,628
Non-trainable params: 0
Total mult-adds (Units.GIGABYTES): 53.95
=====

Input size (MB): 3.15
Forward/backward pass size (MB): 1008.93
Params size (MB): 23.16
Estimated Total Size (MB): 1035.24
=====
```

4b.2 Train (75 Epochs)

```
In [29]: cifar_optimizer_deep_bn = Adam(cifar_model_deep_bn.parameters(), lr=0.001, weight_decay=1e-4)
cifar_module_deep_bn = SimpleModule.classification(cifar_model_deep_bn, num_classes=100, optimizer=cifar_optimizer_deep_bn)
cifar_logger_deep_bn = CSVLogger(LOG_DIR, name="CIFAR100_P4b_deep_bn")

checkpoint_p4b = ModelCheckpoint(monitor="valid_accuracy", mode="max", save_top_k=1, filename="best-p4b")

cifar_trainer_deep_bn = Trainer(
    deterministic=True, max_epochs=75, accelerator="gpu", devices=1,
    logger=cifar_logger_deep_bn,
    callbacks=[ErrorTracker(), TQDMProgressBar(refresh_rate=1), PrintMetrics(), checkpoint_p4b]
)
```

```
cifar_trainer_deep_bn.fit(cifar_module_deep_bn, datamodule=cifar_dm_aug)
```

```
INFO:pytorch_lightning.utilities.rank_zero:GPU available: True (cuda), used: True
INFO:pytorch_lightning.utilities.rank_zero:TPU available: False, using: 0 TPU cores
INFO:pytorch_lightning.utilities.rank_zero: Tip: For seamless cloud logging and experiment tracking,
try installing [litlogger](https://pypi.org/project/litlogger/) to enable LitLogger, which logs metric
s and artifacts automatically to the Lightning Experiments platform.
INFO:pytorch_lightning.accelerators.cuda:LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]
```

	Name	Type	Params	Mode	FLOPs
0	model	CIFARDeepModelBN	5.8 M	train	0
1	loss	CrossEntropyLoss	0	train	0

Trainable params: 5.8 M

Non-trainable params: 0

Total params: 5.8 M

Total estimated model params size (MB): 23

Modules in train mode: 45

Modules in eval mode: 0

Total FLOPs: 0

```
Sanity Checking: |           | 0/? [00:00<?, ?it/s]
Training: |           | 0/? [00:00<?, ?it/s]
Validation: |           | 0/? [00:00<?, ?it/s]
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 1 | train_loss: 3.9400 | train_acc: 0.0906 | valid_loss: 3.1602 | valid_acc: 0.2225 | lr: 0.00
1000
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 2 | train_loss: 3.3223 | train_acc: 0.1837 | valid_loss: 2.7524 | valid_acc: 0.2928 | lr: 0.00
1000
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 3 | train_loss: 2.9259 | train_acc: 0.2548 | valid_loss: 2.3680 | valid_acc: 0.3683 | lr: 0.00
1000
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 4 | train_loss: 2.6061 | train_acc: 0.3171 | valid_loss: 2.3284 | valid_acc: 0.3802 | lr: 0.00
1000
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 5 | train_loss: 2.3783 | train_acc: 0.3653 | valid_loss: 2.2040 | valid_acc: 0.4185 | lr: 0.00
1000
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 6 | train_loss: 2.2100 | train_acc: 0.4034 | valid_loss: 2.0994 | valid_acc: 0.4325 | lr: 0.00
1000
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 7 | train_loss: 2.0754 | train_acc: 0.4353 | valid_loss: 2.2843 | valid_acc: 0.4107 | lr: 0.00
1000
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 8 | train_loss: 1.9566 | train_acc: 0.4645 | valid_loss: 1.9982 | valid_acc: 0.4693 | lr: 0.00
1000
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 9 | train_loss: 1.8523 | train_acc: 0.4864 | valid_loss: 1.9196 | valid_acc: 0.4902 | lr: 0.00
1000
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 10 | train_loss: 1.7824 | train_acc: 0.5022 | valid_loss: 1.9164 | valid_acc: 0.4938 | lr: 0.00
1000
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 11 | train_loss: 1.6906 | train_acc: 0.5239 | valid_loss: 1.6938 | valid_acc: 0.5330 | lr: 0.00
1000
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 12 | train_loss: 1.6354 | train_acc: 0.5380 | valid_loss: 1.7462 | valid_acc: 0.5295 | lr: 0.00
1000
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 13 | train_loss: 1.5699 | train_acc: 0.5561 | valid_loss: 1.5648 | valid_acc: 0.5650 | lr: 0.00
1000
Validation: |           | 0/? [00:00<?, ?it/s]
Epoch 14 | train_loss: 1.5117 | train_acc: 0.5682 | valid_loss: 1.5837 | valid_acc: 0.5571 | lr: 0.00
1000
Validation: |           | 0/? [00:00<?, ?it/s]
```

Epoch 15 | train_loss: 1.4584 | train_acc: 0.5812 | valid_loss: 1.5448 | valid_acc: 0.5706 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 16 | train_loss: 1.4040 | train_acc: 0.5965 | valid_loss: 1.5972 | valid_acc: 0.5655 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 17 | train_loss: 1.3665 | train_acc: 0.6046 | valid_loss: 1.4873 | valid_acc: 0.5892 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 18 | train_loss: 1.3282 | train_acc: 0.6148 | valid_loss: 1.5006 | valid_acc: 0.5886 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 19 | train_loss: 1.2852 | train_acc: 0.6258 | valid_loss: 1.4901 | valid_acc: 0.5836 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 20 | train_loss: 1.2394 | train_acc: 0.6403 | valid_loss: 1.6025 | valid_acc: 0.5729 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 21 | train_loss: 1.2142 | train_acc: 0.6450 | valid_loss: 1.6417 | valid_acc: 0.5637 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 22 | train_loss: 1.1764 | train_acc: 0.6520 | valid_loss: 1.4423 | valid_acc: 0.6003 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 23 | train_loss: 1.1431 | train_acc: 0.6632 | valid_loss: 1.4533 | valid_acc: 0.6047 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 24 | train_loss: 1.1153 | train_acc: 0.6727 | valid_loss: 1.4242 | valid_acc: 0.6130 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 25 | train_loss: 1.0809 | train_acc: 0.6809 | valid_loss: 1.4463 | valid_acc: 0.6098 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 26 | train_loss: 1.0488 | train_acc: 0.6874 | valid_loss: 1.4450 | valid_acc: 0.6109 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 27 | train_loss: 1.0239 | train_acc: 0.6925 | valid_loss: 1.3730 | valid_acc: 0.6267 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 28 | train_loss: 1.0045 | train_acc: 0.6999 | valid_loss: 1.4746 | valid_acc: 0.6098 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 29 | train_loss: 0.9671 | train_acc: 0.7085 | valid_loss: 1.3954 | valid_acc: 0.6258 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 30 | train_loss: 0.9578 | train_acc: 0.7131 | valid_loss: 1.3789 | valid_acc: 0.6303 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 31 | train_loss: 0.9339 | train_acc: 0.7189 | valid_loss: 1.3522 | valid_acc: 0.6423 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 32 | train_loss: 0.8968 | train_acc: 0.7314 | valid_loss: 1.4207 | valid_acc: 0.6233 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 33 | train_loss: 0.8859 | train_acc: 0.7315 | valid_loss: 1.3890 | valid_acc: 0.6339 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 34 | train_loss: 0.8653 | train_acc: 0.7353 | valid_loss: 1.3570 | valid_acc: 0.6420 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 35 | train_loss: 0.8535 | train_acc: 0.7404 | valid_loss: 1.3367 | valid_acc: 0.6417 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 36 | train_loss: 0.8275 | train_acc: 0.7466 | valid_loss: 1.3160 | valid_acc: 0.6484 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 37 | train_loss: 0.8113 | train_acc: 0.7519 | valid_loss: 1.2902 | valid_acc: 0.6523 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 38 | train_loss: 0.7997 | train_acc: 0.7546 | valid_loss: 1.3467 | valid_acc: 0.6440 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 39 | train_loss: 0.7809 | train_acc: 0.7602 | valid_loss: 1.3487 | valid_acc: 0.6430 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]

```
Epoch 40 | train_loss: 0.7676 | train_acc: 0.7648 | valid_loss: 1.2912 | valid_acc: 0.6527 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 41 | train_loss: 0.7483 | train_acc: 0.7692 | valid_loss: 1.3306 | valid_acc: 0.6593 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 42 | train_loss: 0.7303 | train_acc: 0.7757 | valid_loss: 1.4627 | valid_acc: 0.6365 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 43 | train_loss: 0.7278 | train_acc: 0.7751 | valid_loss: 1.3360 | valid_acc: 0.6552 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 44 | train_loss: 0.7147 | train_acc: 0.7785 | valid_loss: 1.3649 | valid_acc: 0.6490 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 45 | train_loss: 0.6906 | train_acc: 0.7858 | valid_loss: 1.3564 | valid_acc: 0.6514 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 46 | train_loss: 0.6934 | train_acc: 0.7850 | valid_loss: 1.3487 | valid_acc: 0.6562 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 47 | train_loss: 0.6677 | train_acc: 0.7932 | valid_loss: 1.3391 | valid_acc: 0.6558 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 48 | train_loss: 0.6686 | train_acc: 0.7926 | valid_loss: 1.3366 | valid_acc: 0.6595 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 49 | train_loss: 0.6531 | train_acc: 0.7975 | valid_loss: 1.3670 | valid_acc: 0.6545 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 50 | train_loss: 0.6468 | train_acc: 0.7990 | valid_loss: 1.4361 | valid_acc: 0.6472 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 51 | train_loss: 0.6277 | train_acc: 0.8056 | valid_loss: 1.3973 | valid_acc: 0.6538 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 52 | train_loss: 0.6273 | train_acc: 0.8069 | valid_loss: 1.3842 | valid_acc: 0.6586 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 53 | train_loss: 0.6053 | train_acc: 0.8123 | valid_loss: 1.3539 | valid_acc: 0.6607 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 54 | train_loss: 0.5983 | train_acc: 0.8138 | valid_loss: 1.3728 | valid_acc: 0.6548 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 55 | train_loss: 0.5937 | train_acc: 0.8136 | valid_loss: 1.4698 | valid_acc: 0.6437 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 56 | train_loss: 0.6002 | train_acc: 0.8131 | valid_loss: 1.3657 | valid_acc: 0.6596 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 57 | train_loss: 0.5809 | train_acc: 0.8188 | valid_loss: 1.4552 | valid_acc: 0.6483 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 58 | train_loss: 0.5742 | train_acc: 0.8221 | valid_loss: 1.3827 | valid_acc: 0.6617 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 59 | train_loss: 0.5755 | train_acc: 0.8198 | valid_loss: 1.3367 | valid_acc: 0.6713 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 60 | train_loss: 0.5670 | train_acc: 0.8216 | valid_loss: 1.3649 | valid_acc: 0.6634 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 61 | train_loss: 0.5541 | train_acc: 0.8266 | valid_loss: 1.3883 | valid_acc: 0.6616 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 62 | train_loss: 0.5507 | train_acc: 0.8284 | valid_loss: 1.3949 | valid_acc: 0.6608 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 63 | train_loss: 0.5351 | train_acc: 0.8335 | valid_loss: 1.4253 | valid_acc: 0.6516 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 64 | train_loss: 0.5455 | train_acc: 0.8295 | valid_loss: 1.3769 | valid_acc: 0.6656 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
```

```

Epoch 65 | train_loss: 0.5398 | train_acc: 0.8310 | valid_loss: 1.3810 | valid_acc: 0.6658 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 66 | train_loss: 0.5222 | train_acc: 0.8362 | valid_loss: 1.3705 | valid_acc: 0.6658 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 67 | train_loss: 0.5295 | train_acc: 0.8336 | valid_loss: 1.4207 | valid_acc: 0.6559 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 68 | train_loss: 0.5201 | train_acc: 0.8368 | valid_loss: 1.4138 | valid_acc: 0.6609 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 69 | train_loss: 0.5122 | train_acc: 0.8407 | valid_loss: 1.4248 | valid_acc: 0.6565 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 70 | train_loss: 0.5064 | train_acc: 0.8417 | valid_loss: 1.5011 | valid_acc: 0.6443 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 71 | train_loss: 0.5101 | train_acc: 0.8431 | valid_loss: 1.4125 | valid_acc: 0.6635 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 72 | train_loss: 0.4921 | train_acc: 0.8461 | valid_loss: 1.4433 | valid_acc: 0.6524 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
Epoch 73 | train_loss: 0.4999 | train_acc: 0.8450 | valid_loss: 1.4218 | valid_acc: 0.6605 | lr: 0.00
1000
Validation: | | 0/? [00:00<?, ?it/s]
INFO:pytorch_lightning.utilities.rank_zero:`Trainer.fit` stopped: `max_epochs=75` reached.
Epoch 74 | train_loss: 0.4932 | train_acc: 0.8453 | valid_loss: 1.3929 | valid_acc: 0.6639 | lr: 0.00
1000

```

4b.3 Plot and Report

```

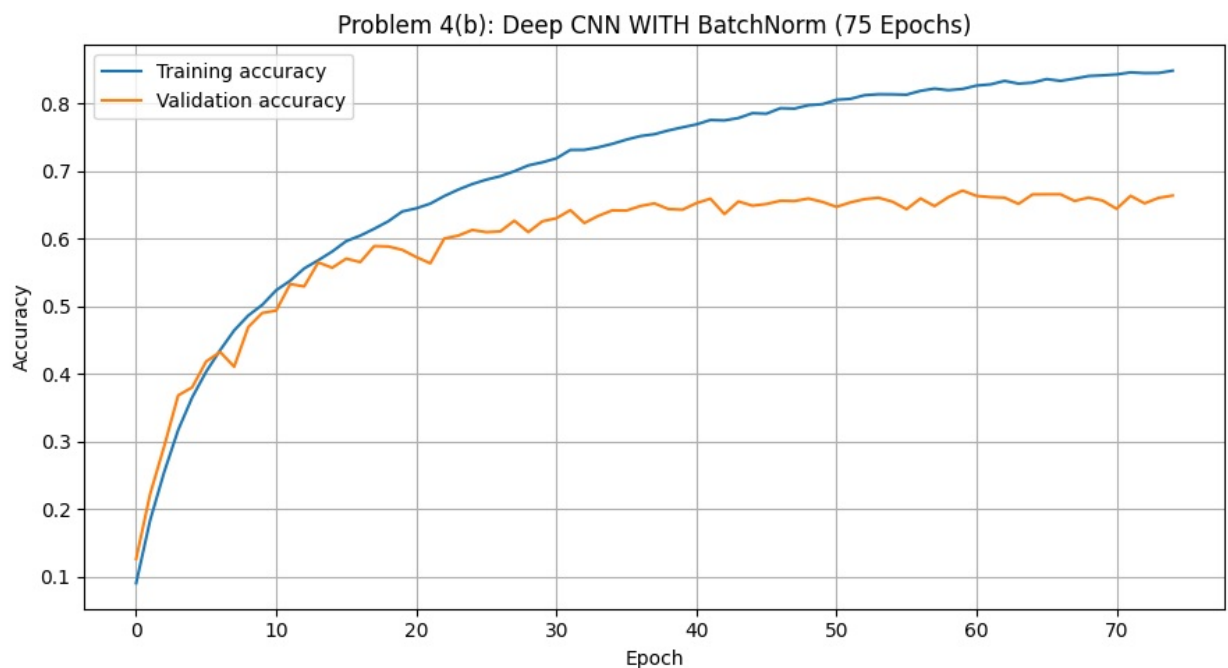
In [30]: epoch_metrics_deep_bn = read_epoch_metrics(cifar_logger_deep_bn)
best_epoch_deep_bn = epoch_metrics_deep_bn["valid_accuracy"].idxmax()
best_valid_acc_deep_bn = epoch_metrics_deep_bn.loc[best_epoch_deep_bn, "valid_accuracy"]
print(f"Best validation accuracy: {best_valid_acc_deep_bn:.4f} at epoch {best_epoch_deep_bn}")

plot_accuracy(epoch_metrics_deep_bn, "Problem 4(b): Deep CNN WITH BatchNorm (75 Epochs)")

test_results_deep_bn = cifar_trainer_deep_bn.test(cifar_module_deep_bn, datamodule=cifar_dm_aug, c
problem4b_test_accuracy = test_results_deep_bn[0]["test_accuracy"]
print(f"Problem 4(b) test accuracy: {problem4b_test_accuracy:.4f}")
print(f"Problem 4(b) test error: {1-problem4b_test_accuracy:.4f}")

```

Best validation accuracy: 0.6713 at epoch 59



```
INFO:pytorch_lightning.utilities.rank_zero:Restoring states from the checkpoint path at /content/drive/MyDrive/hw7_logs/CIFAR100_P4b_deep_bn/version_0/checkpoints/best-p4b.ckpt
INFO:pytorch_lightning.accelerators.cuda:LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]
INFO:pytorch_lightning.utilities.rank_zero:Loaded model weights from the checkpoint at /content/drive/MyDrive/hw7_logs/CIFAR100_P4b_deep_bn/version_0/checkpoints/best-p4b.ckpt
/usr/local/lib/python3.12/dist-packages/pytorch_lightning/utilities/_pytree.py:21: `isinstance(treespec, LeafSpec)` is deprecated, use `isinstance(treespec, TreeSpec)` and `treespec.is_leaf()` instead.
Testing: |          | 0/? [00:00<?, ?it/s]
Problem 4(b) test accuracy: 0.6652
Problem 4(b) test error: 0.3348
```

Problem 4(b) Interpretation

Adding batch normalization — `nn.BatchNorm2d` after each convolutional layer and `nn.BatchNorm1d` before the dense ReLU — dramatically improved training compared to Problem 4(a). By normalizing activations within each mini-batch at every layer, batch norm keeps the inputs to each ReLU in a healthy range, maintaining gradient flow through the deeper network and preventing the saturation and dead-neuron problems that limited Problem 4(a).

The results were the strongest of the entire assignment: the best validation accuracy reached 0.6713 at epoch 59, and the test accuracy was 0.6652 (test error: 0.3348) — an improvement of over 16 percentage points compared to Problem 4(a), and the highest test accuracy across all four problems. The training and validation curves also tracked each other more closely than in 4(a), indicating that batch normalization helped not only with optimization but also with generalization. This confirms that batch normalization is essential when going deeper than a single convolutional layer per block.

Summary: Test-Set Results for All Models

```
In [31]: results_table = pd.DataFrame([
    {"Model": "P1: Original CNN (50 epochs)", "Test Accuracy": best_model_test_accuracy},
    {"Model": "P2: Original CNN + Augmentation (300 epochs)", "Test Accuracy": problem2_test_accuracy},
    {"Model": "P3: Wider CNN + Augmentation (300 epochs)", "Test Accuracy": problem3_test_accuracy},
    {"Model": "P4a: Deep CNN, No BatchNorm (300 epochs)", "Test Accuracy": problem4a_test_accuracy},
    {"Model": "P4b: Deep CNN + BatchNorm (300 epochs)", "Test Accuracy": problem4b_test_accuracy}
])

results_table["Test Accuracy"] = results_table["Test Accuracy"].map("{:.4f}".format)
results_table["Test Error"] = results_table["Test Error"].map("{:.4f}".format)
display(results_table)
```

	Model	Test Accuracy	Test Error
0	P1: Original CNN (50 epochs)	0.5189	0.4811
1	P2: Original CNN + Augmentation (300 epochs)	0.5754	0.4246
2	P3: Wider CNN + Augmentation (300 epochs)	0.6134	0.3866
3	P4a: Deep CNN, No BatchNorm (300 epochs)	0.4988	0.5012
4	P4b: Deep CNN + BatchNorm (300 epochs)	0.6652	0.3348

Final Discussion

The results across the four problems show a clear progression in test accuracy as each architectural and data improvement was applied:

- **P1 (Original CNN, 25 epochs):** test accuracy **0.5189** — the baseline showed clear overfitting, with training accuracy reaching 0.79 while validation plateaued around 0.52.
- **P2 (Augmentation, 75 epochs):** test accuracy **0.5754** — data augmentation reduced overfitting and improved generalization by +5.6 percentage points.
- **P3 (Wider network, 75 epochs):** test accuracy **0.6134** — doubling the channel width gave the model more capacity and pushed accuracy up by another +3.8 points.
- **P4a (Deep, no BatchNorm, 40 epochs):** test accuracy **0.4988** — adding depth without normalization hurt performance, dropping below even Problem 2, demonstrating the gradient flow problem in deeper

unnormalized networks.

- **P4b (Deep + BatchNorm, 75 epochs):** test accuracy **0.6652** — batch normalization resolved the training instability, producing the best result overall and improving over Problem 3 by +5.2 percentage points.

The key takeaway is that architectural depth alone is insufficient — normalization is what makes deeper networks trainable in practice. Combined with data augmentation and increased width, batch normalization enabled the deepest model to achieve the best generalization.